

Beneficiaries' Preferences for the Design and Targeting of Humanitarian Food Assistance: Evidence from DRC and Afghanistan*

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Abstract

This paper examines beneficiaries' preferences over humanitarian food assistance and how the constraints that define fragility shape them. We implement a conjoint experiment in the Democratic Republic of the Congo (DRC) and Afghanistan to elicit preferences over the modality, frequency, intra-household recipient, and targeting method of assistance. Delivery modality dominates in both settings. In the DRC, beneficiaries would forgo more than the entire monthly transfer for cash in hand rather than in-kind food, while mobile money carries a penalty of about 30% of the transfer. Modality valuations are an order of magnitude smaller in Afghanistan. Preferences over other attributes differ across settings. Designating a woman as recipient is worth about 48% of the transfer in the DRC but is dispreferred in Kabul and Mazar-i-Sharif. One finding holds across contexts. Community-based targeting is dispreferred in all three sites, with a penalty of about 69% of the transfer in the DRC. Randomized exposure to different targeting methods and delivery modalities moves no individual valuation: in neither country does the assigned arm change what respondents will give up for any single attribute. For agencies facing funding cuts, these estimates identify changes in program design that could help maintain beneficiaries' utility despite smaller transfers.

Keywords: humanitarian assistance, program design, discrete choice experiment, willingness to pay, targeting governance, DRC, Afghanistan

JEL Codes: O12, D12, H53, I38

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1 Introduction

Humanitarian systems face mounting fiscal pressure. Global needs continue to expand while aid budgets contract (FSIN and GNAFC, 2025; UN OCHA, 2024), forcing policymakers to decide how to allocate scarce resources. One common margin of adjustment has been to reduce the size of the transfer to cope with funding cuts.¹ The cost of assistance, however, depends not only on the value of the transfer but also on other features of its design, such as the delivery modality or the frequency with which assistance is provided. A more recent margin of adjustment therefore examines the structure of aid packages itself, seeking configurations that deliver greater impact per dollar. This has led to a growing body of evidence showing that such design choices generate significant differences in welfare outcomes.² Yet this evidence has been concerned primarily with measuring downstream outcomes and costs, with little insight into beneficiaries’ preferences over such configurations and the utility they derive from them, the constraints shaping those preferences, or how exposure to these interventions may itself alter those preferences.

In this paper, we study beneficiary preferences over the design of humanitarian assistance through a cross-country conjoint experiment conducted among beneficiaries of World Food Programme (WFP) in the Democratic Republic of the Congo (DRC) and Afghanistan. The

¹Evidence on the effects of cutting assistance, rather than ending it, is still thin. Sterck and Bruni (2025) exploit variation in survey timing around a 20% cut to cash assistance for refugees in Kenya and find large declines in food security and expenditure; the same variation applied to disbursement delays yields comparable losses (Bruni and Sterck, 2025). Iqbal et al. (2026) instead find no adverse effects of unanticipated program exit in Pakistan, using a regression discontinuity around a revised eligibility threshold. Evidence from high-income countries, where administrative data allow long panels, points to short-lived labor-supply responses but persistent costs to children’s human capital (Dustmann et al., 2024a,b). In humanitarian settings the projected consequences of cuts come largely from simulation (WFP, 2023).

²Three design margins have received the most attention. First, the choice of modality — cash, vouchers, or in-kind aid — has a long literature, from theoretical syntheses (Currie and Gahvari, 2008) to experimental tests of paternalism and price effects (Cunha, 2014; Cunha et al., 2019) and reviews of the humanitarian “cash versus food” debate (Gentilini, 2016); in fragile and humanitarian settings it has been compared in the DRC (Aker, 2017), Niger (Hoddinott et al., 2018), and Yemen (Schwab, 2020). Second, the tranching of assistance (how a fixed transfer value is split across payments) is examined by Kansikas et al. (2023), who tests the frequency of cash transfers, and Haushofer and Shapiro (2016), who jointly randomises transfer size, frequency, and recipient gender. Third, and more recently, the timing of assistance, in particular evidence on anticipatory action (Gros et al., 2020; Derbyshire et al., 2025). For more evidence of interventions in humanitarian settings, see Jeong and Trako (2022).

experiment elicits preferences over central program attributes—delivery modality, transfer frequency, the intra-household recipient, and the targeting mechanism—in settings where aid constitutes one of the main sources of household income and where beneficiaries face limited access to credit and labor markets (Banerjee and Duflo, 2007).

Beyond its limited scope, the literature on beneficiaries’ preferences over assistance features exhibits several limitations that this paper addresses by exploiting a distinctive combination of setting, data, and study design. First, the existing evidence draws largely from stable settings, where the features that define fragile contexts are far less salient.³ These features include economic frictions such as limited liquidity and credit, weak institutional capacity, weakened social ties, and recurrent exposure to shocks (Banerjee and Duflo, 2007; Mullainathan and Shafir, 2013; Schilbach et al., 2016). Here, we work in two of the largest humanitarian emergencies in the world, where widespread poverty, protracted conflict, and recurrent shocks have left an estimated 25.6 million people in the DRC and 14.8 million in Afghanistan facing high levels of acute food insecurity (IPC Phase 3 or above) (FSIN and GNAFC, 2025). Running an identical design across two distinct crises allows us to test whether preferences are context-specific or travel across fragile settings.

Second, existing evidence offers little insight into how preferences vary across beneficiaries or with the characteristics and constraints that define fragility. We address this by embedding the conjoint within two large WFP impact evaluations, which yield a large sample and a rich set of household variables. These data let us go beyond average and context-specific preferences and examine how valuations over specific attributes of assistance vary both with household characteristics, such as educational attainment and food security, and with the frictions and shocks that define fragility, such as displacement, conflict exposure, and adverse market conditions.

³Existing preference evidence comes mostly from less fragile contexts and development programs: conditional cash transfer programs in India (Rao et al., 2020) and Ghana (Yeboah et al., 2019), and the Ghana School Feeding Programme (Abreh et al., 2024). Evidence from fragile settings is scarcer, and each study focuses on a single attribute: cash versus in-kind payments under Ethiopia’s Productive Safety Net Programme (Hirvonen and Hoddinott, 2021), modality in Sudan (Abushama et al., 2025), and preferences over who should be in charge of targeting decisions in Haiti (López García and Berens, 2026).

Assistance is itself a bundle of design decisions, yet an understudied aspect of preferences is how beneficiaries value these features and trade them off against one another, rather than one dimension at a time, as is mostly the case in the existing literature.⁴ Our conjoint design jointly values these attributes. Alongside them it varies the value of the transfer itself, at 17%, 33%, and 50% below the status quo amount each household receives under its current WFP program. That variation is what makes the exercise quantitative rather than ordinal: because respondents trade design features against the size of the transfer, every attribute can be expressed in money-metric terms, as the share of the transfer a household would give up to obtain it. Preferences over program design and the value of the transfer are thereby placed on a common scale, which is what allows design choices to be compared against one another and, in Section 6.7, against reductions in program budgets. This is of immediate policy relevance, as both operations are among the most severely underfunded in the world, with funding gaps of 85% (US\$1.46 billion) in Afghanistan and 66% (US\$925 million) in the DRC by 2026 (OCHA, 2026a,b).

The embedding of the conjoint experiments within an impact evaluation provides a rare opportunity to causally examine how exposure to changes in transfer modality and targeting shapes beneficiaries’ preferences.⁵ We exploit the randomized design of both impact evaluations, which varied the modality of assistance in Afghanistan and the targeting method in the DRC, to compare stated preferences with those of households randomly exposed to a different delivery modality or targeting approach. In the DRC, where the evaluation randomized communities into a new community-driven approach, we go one step further and examine preferences over committee governance: we administered a second conjoint, to the

⁴A small conjoint experiment literature elicits beneficiaries’ joint preferences over design attributes, but it is concentrated on conditional cash transfer (CCT) programs: Abreh et al. (2024) on the Ghana School Feeding Program, Yeboah et al. (2019) on CCT design in Ghana, and Rao et al. (2020) on a maternal- and child-health CCT in India. To our knowledge, comparable cross-country evidence for unconditional humanitarian assistance is absent, and the simulation of aid cuts in a conjoint design has so far only been used by Garcia et al. (2026), but for a bundled resilience program among subsistence farmers in the South Kivu, DRC.

⁵Shapiro (2019) and Hidrobo et al. (2014) conduct a comparable causal test of how exposure to a transfer modality shapes beneficiaries’ preferences, but in stable development settings rather than in fragile, humanitarian ones.

same respondents, on the gender composition of selection committees, how members are chosen, and how they reach decisions on beneficiary selection.

Lastly, we use the rich household data to examine which household characteristics predict a preference for more liquid assistance. We classify assistance packages according to their degree of liquidity and pool data across all locations to identify the household characteristics associated with different choices.

In the DRC sample, we find that beneficiaries' utility loads heavily on the non-monetary features of the package. Respondents would forgo more than the full transfer to receive cash in hand rather than the in-kind food parcel used as the reference modality (willingness to pay 169% of the status quo transfer; Section 6.2 sets out how much weight that particular ratio can bear). This is an especially policy-relevant finding given that, in 2024, in-kind food still reached roughly 78% of WFP's beneficiaries in the DRC, against only 22% for cash-based transfers (WFP, 2024), even though both WFP's own operations and other organizations active in the country⁶ have shown that mobile money and other cash modalities are available and feasible at scale. Designating a woman as recipient commands a premium of 48% of the status quo transfer, while community-based targeting is not a premium at all but a substantial penalty (−69%): beneficiaries prefer selection by data to selection by a community committee, and households the evaluation randomly assigned to community-based targeting disprefer it almost as strongly as those never exposed to it. Bi-monthly delivery is modestly preferred to monthly in the DRC (14%), though this is one of several attributes whose sign reverses across sites, while mobile money is disliked relative to the in-kind reference (−30%). All four delivery and procedural attributes display substantial preference heterogeneity, much of it correlated with food security, education, and prior exposure to community-based targeting.

In Afghanistan the picture differs on almost every margin. Cash in hand remains pre-

⁶For instance, GiveDirectly delivers mobile-money cash transfers in the DRC (<https://www.givedirectly.org/drc>), and the World Bank supported a digital cash-transfer response in Kinshasa during the COVID-19 pandemic (<https://www.brookings.edu/articles/cash-and-the-city-digital-covid-19-social-response-in-kinshasa/>).

ferred to the site-specific alternative in both cities, but the premium is far smaller than in the DRC, and it depends on what it is measured against. In Mazar, where the omitted modality is a voucher, cash still commands a clear premium. In Kabul, where the omitted modality is a bank deposit, the premium all but disappears. That is what one would expect if what beneficiaries value is liquidity rather than cash as such: a bank deposit is already a close substitute for cash in hand, an in-kind food parcel is not. Mobile money follows the same logic in reverse. In Kabul it is indistinguishable from a bank deposit, which is to say that respondents there see no difference between two forms of electronic money. In Mazar, measured against a commodity voucher, it is strongly dispreferred. Afghan respondents are also markedly more sensitive to the size of the transfer itself: the utility cost of a 50% reduction is roughly four times as large in Kabul and Mazar as it is in the DRC.

Two attributes reverse outright. Designating a woman as recipient, which DRC respondents value highly, is a penalty on average in both Afghan sites. That contrast is smaller than it appears, and Section 6.3 takes it apart: the Afghan average is driven by male respondents and by households in which women cannot travel without permission, while Afghan women who answer for themselves value a female recipient positively. Since the DRC sample is female by design and the Afghan samples are not, part of the reversal is a difference in who was asked rather than in where they live. A programme designing around intra-household targeting would need to take that distinction seriously, because the two readings imply different things about whose preference a transfer rule would be accommodating. Bi-monthly delivery, mildly preferred in the DRC, is dispreferred in Kabul (-0.25) and heavily so in Mazar (-0.85). Only community-based targeting behaves the same way everywhere: it is a penalty in Kabul (-0.40) and Mazar (-1.41) as it is in the DRC, and it is the one attribute whose sign is common to all three sites.

Contrasting the two contexts, the striking result is how little transports. A pooled model with site $\times$ attribute interactions rejects equality of valuations across the three sites for every attribute we measure ($\chi^2(16) = 1219.2$, $p = 0.001$), and every attribute-specific test rejects at

$p < 0.001$; Appendix Table A6). The common finding is the objection to community-based selection; almost everything else is context-specific, and two attributes do not merely differ in magnitude but change sign.

Two qualifications belong with that claim. The omitted modality is a different good in each site, so part of the divergence in the modality coefficients reflects the baseline rather than the preference — indeed the Kabul result is most naturally read that way. And logit coefficients are identified only up to the scale of the error term, so a rejection of equal coefficients is a rejection of equal preferences only if scale is common across sites. We settle that question rather than assume it away: willingness to pay is a ratio of two coefficients from the same sample, so scale cancels, and the by-site willingness-to-pay estimates in Section 6.2 still differ by an order of magnitude and still reverse sign. Neither qualification accounts for a sign reversal, which is why we read the recipient and frequency results as genuine differences in preference.

The most consequential implication is that pooling can manufacture a null. The pooled recipient-gender coefficient is 0.02 and statistically indistinguishable from zero. That is not an absence of preference: it is a strong DRC positive averaged against two Afghan negatives, and it describes no site in the study. Multi-country evidence on assistance design that reports pooled coefficients should be read with that possibility in mind. Whether a stated preference merely reflects prior exposure to the mechanism is a question both evaluations let us answer, on different margins: the DRC randomised the targeting method and the Afghan sites randomised the delivery modality. Neither randomisation moves the corresponding preference (Section 6.4).

To better situate our results, we revisit the existing evidence on beneficiary preferences in the following section.

2 Preferences over Assistance Design and Targeting

2.1 Modality

The most studied preference regarding the design of assistance is the delivery modality. Three factors are commonly identified as shaping the choice between modalities, particularly between in-kind and cash transfers. First, the inframarginality of the assistance: giving in-kind aid rather than cash affects a household’s consumption only if the transfer exceeds what the household would have spent normally on food; below that point, cash and in-kind are effectively equivalent (Cunha, 2014). Second, behavioral factors, such as self-control problems, present bias, risk preference, and intra-household bargaining over who controls the transfer (Schilbach et al., 2016; Mullainathan and Shafir, 2013; Haushofer and Shapiro, 2016). Third, contextual factors, including market access, inflation concerns, and local food availability (Gentilini, 2016; Hirvonen and Hoddinott, 2021).

These three factors -inframarginality, behavioral and contextual factors- are likely to be more acute in a humanitarian context. For instance, the value of the transfer may represent a larger share of the household’s income than in other social protection or anti-poverty programs. Likewise, price and market concerns are likely more pronounced in conflict or poly-crisis settings, characterized by recurrent shocks that are also more likely to intensify behavioral responses such as present bias and self-control problems among beneficiaries.

Evidence on modality preferences from more stable settings is mixed. In Ecuador, Hidrobo et al. (2014) find that recipients place substantial value on the flexibility of cash, with cash beneficiaries reporting higher satisfaction with their modality (77%) than in-kind (55%) or voucher (56%) recipients. In Kenya, by contrast, Shapiro (2019) finds a clear preference for in-kind programs over cash: when asked their indifference point between cash and three common development programs (agricultural extension, inputs, and poultry), beneficiaries valued the programs at two to five times their cash-equivalent cost, indicating that most would forgo a substantial share of the transfer’s monetary value to receive it in-kind.

A preference for in-kind transfers also emerges from studies in Nepal and India, even where cash programs function well. Khanal et al. (2024), in an incentivized experiment among Nepalese farmers, find that roughly 67% chose in-kind transfers over cash of equal value. However, among those initially choosing cash, 52% kept that choice even when offered in-kind goods worth up to 45% more, echoing the broader pattern that beneficiaries tend to favor the modality they know. Ghatak et al. (2016) report that 55% of beneficiaries of Bihar’s bicycle scheme would have preferred receiving a bicycle directly rather than the cash transfer. Abreh et al. (2024) report preference for in-kind in Ghana, where on-site school meals were preferred over both cash transfers and take-home rations, particularly among households vulnerable to food price volatility and food insecurity.

Modality preferences vary systematically with region of residence. In Kenya, rural households and those in high-price areas lean toward in-kind modalities, a pattern that likely reflects market access (Michelson et al., 2012; Shapiro, 2019). In Nepal, each additional ten minutes of travel time to the nearest market raises the probability of preferring in-kind by roughly two percentage points (Khanal et al., 2024), and in Bihar each additional kilometer to a bicycle store reduces the probability of preferring cash by 1.2 percentage points (Ghatak et al., 2016). Market conditions also appear to interact with the gender of the household head: female-headed households and those in rural areas are particularly likely to prefer in-kind transfers (Michelson et al., 2012; Shapiro, 2019). Income and liquidity constraints operate in the same direction. Poorer households and those without savings are more likely to prefer in-kind transfers (Abreh et al., 2024; Das and Sethi, 2023), particularly when the cash amount is insufficient to purchase the intended good without borrowing or drawing on savings (Ghatak et al., 2016; Khanal et al., 2024).

Beyond the heterogeneities by gender, socioeconomic status, and market conditions, two further points emerge clearly from these settings. First, the type of in-kind good matters: bicycles, fertilizer, and other agricultural inputs are consistently valued more highly than their cash equivalents. Second, preferences appear to be neither shaped by program effective-

ness nor predictive of welfare outcomes. In India, bicycles were preferred over cash despite the cash program performing well on coverage, leakage, and grievance indicators (Ghatak et al., 2016), and in Kenya, Shapiro (2019) finds that beneficiaries randomly assigned to their preferred modality fared no better than those assigned to the alternative, suggesting that stated preferences do not identify who would benefit most from each transfer type.

Evidence from humanitarian contexts is also heterogeneous. Syrian refugee households display a broad preference for cash (Doocy et al., 2016), and in the Democratic Republic of the Congo, cash transfers are both more cost-effective than in-kind assistance and, qualitative evidence suggests, preferred by beneficiaries (Aker, 2017). Abushama et al. (2025) document comparable heterogeneity among internally displaced persons in Sudan, where roughly half of respondents prefer cash and the remainder is divided between in-kind and hybrid transfers. In Ethiopia, by contrast, most beneficiaries prefer food despite the ready availability of cash, with the share preferring cash exclusively never exceeding 17% (Hirvonen and Hoddinott, 2021). This preference also appears to track seasonal conditions. In-kind transfers are favored during the lean season, when food is scarce and prices are high, whereas cash is preferred around the harvest, when food is abundant and cheap (Sabates-Wheeler and Devereux, 2010).

Modality preferences in fragile settings also vary systematically with household characteristics. In Ethiopia, more food-secure households and those with better access to food markets and financial services are more likely to prefer cash, whereas food-insecure households with limited market access tend to prefer food transfers (Hirvonen and Hoddinott, 2021). Abushama et al. (2025) find a similar pattern in Sudan, where limited market access again predicts a stronger preference for in-kind transfers. Beyond market access, their study shows that preferences also track household agency and intra-household bargaining dynamics, consistent with theoretical predictions (Doss, 2013): wives report a preference for in-kind transfers 7 percentage points higher than their husbands. Both studies further identify self-control constraints as an important determinant of the in-kind preference. Implementation

performance also appears to matter, a consideration particularly salient in fragile settings. Collyer et al. (2025) find beneficiaries who perceive the state as performing poorly tend to prefer cash.

In general, evidence directly comparing cash with equivalent in-kind food in development settings is thin. Beneficiaries tend to prefer in-kind transfers when the goods are productive, while the most comprehensive evidence on food specifically, Hidrobo et al. (2014) in Ecuador, finds that cash is preferred over an equivalent food basket. In fragile contexts, by contrast, the evidence leans toward cash, with households facing severe food insecurity and limited market access standing out as the main exception.

A further gap concerns the delivery mechanism. Digital payment systems have been shown to lower delivery costs and improve nutritional outcomes relative to manual cash distribution in crisis-affected settings (Aker et al., 2016), but little is known about beneficiary preferences in this context. In India, beneficiaries endorse digital delivery decisively compared to traditional delivery methods. Muralidharan et al. (2016) find that 90% of NREGS beneficiaries and 93% of social pension beneficiaries in Andhra Pradesh prefer a Smartcard-based system to the status quo, with only 3% in either program disagreeing. In Bangladesh, by contrast, Ambler et al. (2025) show that only 8% of respondents accept mobile money at parity with cash, and the average respondent is willing to forgo 43% of the transfer value to receive physical cash instead. This premium is driven mainly by individual-level frictions such as mobile-phone access, familiarity, and perceived fraud risk. How beneficiary preferences extend to these digital modalities in contexts such as the DRC and Afghanistan, where mobile money has become increasingly central to humanitarian operations, remains underexplored.

2.2 Frequency

Evidence on the temporal structure of assistance focuses mainly on tranching, comparing lump-sum payments with smaller installments. This literature tests whether households

facing financial market frictions use assistance as a financial instrument rather than for immediate consumption (Kansikas et al., 2023), and whether larger transfers are consistent with the asset thresholds implied by poverty-trap models. A transfer delivered as a single lump sum can push a household past the critical threshold beyond which it accumulates capital on its own, whereas smaller installments are absorbed into consumption (Carter and Barrett, 2006; Balboni et al., 2022). Frequency, by contrast, remains under-explored, even though it speaks directly to a distinct dimension of beneficiary welfare: liquidity.

Evidence from broader social protection schemes consistently shows that recipients struggle to spread consumption evenly between infrequent payments. Pension recipients spend more right after each monthly check and cut back toward the end of the month (Stephens Jr., 2003), and those paid monthly smooth food expenditure more effectively and report fewer hunger spells than those paid every two months (Aguila et al., 2017). Food stamp recipients show a parallel pattern in caloric intake (Shapiro, 2005). This end-of-period decline is concentrated among households without savings buffers (Mastrobuoni and Weinberg, 2009), indicating a liquidity problem rather than an income one. Laamanen et al. (2022) find that switching from monthly to twice-monthly payments removes the end-of-month spending drop, and Hoang (2025) shows that workers who struggle to smooth consumption accept lower total pay in exchange for more frequent paychecks. This evidence suggests that beneficiaries with the tightest liquidity constraints should prefer more frequent transfers, an implication that has not been systematically researched for humanitarian assistance.

Evidence on payment frequency in cash transfer programs points to a similar liquidity-management role in less fragile settings. In rural Kenya, Haushofer and Shapiro (2016) find that recipients of monthly unconditional cash transfers in rural Kenya achieved greater food security than those receiving an equivalent lump sum, while lump-sum recipients were more likely to spend on durable assets, suggesting that more frequent payments support consumption smoothing, whereas lump sums are channeled into investment.

Also in Kenya, Kansikas et al. (2023) find that when given control over the structure

of a one-year transfer, almost all recipients chose one or two large tranches rather than monthly installments, with similar patterns documented in Malawi and Liberia. A similar pattern emerges in labor contexts: Brune and Kerwin (2019) find that 74% of agricultural workers in Malawi preferred a deferred lump-sum wage over weekly installments, and those randomly assigned to the lump-sum schedule were 50% more likely to purchase a high-return short-term bond. Both studies interpret these patterns as evidence that liquidity constraints, rather than savings preferences, drive the appetite for deferral.

Preferences toward investment oriented designs appear to maximize welfare in stable settings. In Kansikas et al. (2023), mid-year transfers generated higher subsequent income than immediate ones, with many recipients reporting in retrospect that they would have preferred to defer further. At the same time, the welfare effects of payment structure appear to depend on the size and frequency of the transfer. Brune et al. (2017) find that for a one-time \$60 windfall during the lean season in Malawi, neither a savings default nor a payment delay of up to eight days affected the amount or composition of recipients' subsequent spending, suggesting that poor households can manage modest, infrequent transfers effectively on their own.

2.3 Recipient

The practice of directing transfers to women has been widely adopted across social protection programs worldwide (Peterman et al., 2026; Hidrobo et al., 2026), in line with evidence that doing so generates positive impacts on consumption, nutritional outcomes, and women's empowerment (e.g., Haushofer and Shapiro, 2016; De Brauw et al., 2014; Armand et al., 2020), and raises participation in conditional assistance programs (Balantrapu et al., 2025). This body of evidence, however, focuses mostly on ex post outcomes. The few studies eliciting preferences directly come from middle-income settings: in Argentina, both men and women prefer the transfer to be paid to mothers, citing women's role as primary carers and their economic vulnerability in the event of separation (Rabinovich and Diepeveen,

2015); in Macedonia, women place a positive willingness to pay on receiving the transfer themselves rather than their husbands, and those randomly assigned the transfer score higher on empowerment indices (Almås et al., 2018). Whether similar preferences hold in fragile settings, where extreme poverty and distinct gender and social norms set the context apart, remains an open question.

2.4 Targeting method

Direct evidence on beneficiary preferences over the targeting method itself is thin, and comes mostly in the form of satisfaction or legitimacy measures collected alongside experiments testing different targeting approaches. Alatas et al. (2012) randomize community-based, proxy-means, and hybrid targeting across 640 Indonesian villages. Although PMT slightly outperforms CBT on consumption-based accuracy, beneficiaries report higher satisfaction with CBT-allocated lists, a stated-satisfaction margin that can plausibly be read as revealed preference for the procedure. The result does not generalize. In Niger, Premand and Schnitzer (2021) run a parallel randomization and find the opposite ranking, with beneficiaries perceiving PMT as more legitimate than CBT. In Bangladesh, Aiken et al. (2025) layer phone-based machine-learning targeting on top of CBT in the same villages and document a 27 percentage-point satisfaction premium and a 35 percentage-point fairness premium for CBT over the phone-based rule. They show, however, that this gap is largely driven by which method a household actually received. Recipients of the CBT transfer prefer CBT, and recipients of the phone-based transfer prefer the phone-based rule.

A limited number of studies have attempted to elicit preferences over the selection carried out by community committees. Schüring (2014) runs allocation experiments in rural Zambia and shows that community members are mildly pro-poor when distributing transfers among themselves, with little favoritism beyond a tendency to allocate to those who share their characteristics, and that group decisions are more progressive than individual ones. A contrasting result comes from rural Ethiopia, where Abay et al. (2026) randomize

180 villages to a targeting experiment allowing local committees to retain a share of the transfer budget as administrative costs. They find that group decisions extract higher rents than individual proposals, with collective outcomes disproportionately driven by the most extreme self-interested member rather than by the majority. In both cases, however, the preferences elicited are over selection criteria within a fixed CBT setup, not over the targeting rule or the committee’s design. The governance parameters of CBT — who sits on the committee, gender balance, deliberation process — have been studied for their effects on allocation outcomes (Basurto et al., 2020; Camacho and Conover, 2011; Casey, 2018) but not directly elicited as preferences. Our second experiment fills this gap by recovering beneficiaries’ valuations of committee-design attributes in villages of eastern DRC randomly assigned to a community-targeting rule.

3 Setting and Intervention

3.1 DRC

The DRC has been facing one of the world’s most severe and protracted humanitarian crises. In 2025, an estimated 27.7 million Congolese were acutely food insecure, with 10.3 million in the eastern provinces of North Kivu, South Kivu, Ituri, and Tanganyika, which form the epicenter of the emergency (IPC, 2025a,b). Escalating conflict, mass displacement (5 million internally displaced as of August 2025), and recurrent epidemics of cholera, Mpox, and Ebola have strained food and nutrition security further. An estimated 6.5 million people, mostly children under five and pregnant or breastfeeding women, were identified as in need of acute-malnutrition treatment or prevention (IPC, 2025c).

WFP is the largest humanitarian food-security actor in the DRC, delivering assistance through three main modalities: in-kind food, cash-based transfers, and capacity strengthening. In 2023, WFP reached close to 6 million beneficiaries, with delivery concentrated in its General Food Assistance (GFA) program. Assistance was predominantly in-kind: 3.68

million beneficiaries (61%) received in-kind food, 2.12 million (35%) received cash-based transfers, and 0.19 million (3%) were reached through capacity strengthening (World Food Programme, 2024). By 2025, both the scale and the composition of delivery had changed. Total beneficiaries reached fell to 5.26 million, a 12% reduction, and the modality mix shifted sharply toward in-kind food, whose share of beneficiaries rose from 61% to 82% (4.34 million people, +18% in absolute terms), while cash-based transfers contracted from 2.12 million to 0.90 million (−58%) and their share fell from 35% to 17% (World Food Programme, 2026b).

WFP’s GFA program provides unconditional cash transfers as lean-season support, delivered over three to six months depending on funding availability. This study is embedded in the 2023 roll-out of the GFA program in Tanganyika province, which ran from May to October 2023. Beneficiaries in the study received assistance in four tranches, scaled to household size. The first three payments were 30,000 CDF per household member; the final payment rose to 36,000 CDF (about USD 15 in nominal terms, or 39 PPP dollars⁷) after the country’s Cash Working Group updated the Minimum Expenditure Basket (MEB) to which the GFA transfer value is calibrated. Cumulative transfers over the cycle amount to 126,000 CDF per household member (about USD 54), covering roughly 57% of the expenditure required to meet an average household’s needs over the six-month period.⁸

The choice experiment prices attributes against the transfer level that prevailed for most of the cycle. It held a 30,000 CDF “full transfer” fixed as its top level and varied reductions from it (Section 4.2), so every willingness-to-pay estimate below is a share of 30,000 CDF ($\approx$ USD 12.82) — the amount paid in the first three tranches, and below the 36,000 CDF of the final one.

This roll-out of the GFA was also part of an impact evaluation testing a new community-

⁷PPP-adjusted figures use the World Bank purchasing-power-parity conversion factor for household and NPISH final consumption expenditure, 928.60 CDF per international dollar (World Development Indicators, series PA.NUS.PRVT.PP; <https://data.worldbank.org/indicator/PA.NUS.PRVT.PP?locations=CD>). That series has no DRC observation after 2021, so we use the latest published value. The GDP-based factor PA.NUS.PPP does extend to 2023, but measures the wrong basket for valuing a household transfer.

⁸The coverage ratio divides the cumulative transfer by the six-month per capita expenditure requirement of 221,182 CDF reported in WFP’s Emergency Food Security Assessment (EFSA) for the DRC, 2023.

based targeting (CBT) approach for identifying beneficiaries at the household level.⁹ Conducted in Tanganyika province, the evaluation randomizes 84 blocs¹⁰, nested within 33 villages, to either the new CBT approach or WFP’s standard household-level targeting method, Proxy-Means-Test “Plus” (PMT+). All blocs received the same GFA package; only the procedure for identifying eligible households differed. PMT+ combines a data-driven model of household food insecurity with categorical “lock-in” vulnerability criteria identified through focus groups, while CBT convenes a 12-member committee of bloc residents (six women and six men, reflecting diversity in age, ethnicity, and disability status) that defines the eligibility criteria for its own community. The evaluation found that the two methods performed similarly on targeting precision and welfare outcomes, yet they diverged on coverage: the share of eligible households reached was about 11 percentage points lower under CBT than under PMT+, even though communities under the two procedures faced comparable levels of food insecurity (Dunsch et al., 2026). This coverage gap was mirrored in perceptions of safety, with respondents in CBT blocs reporting a higher likelihood of feeling unsafe.

3.2 Afghanistan

Afghanistan is the second setting, and it differs from the DRC on the dimension that matters most for this paper. The scale of need is of the same order — close to a third of the population faces acute food insecurity, against roughly a quarter in the DRC — but the assistance architecture beneficiaries have experience of is different. Some 13.8 million people are projected to be acutely food insecure in 2026 and 4.9 million women and children expected to require treatment for malnutrition (World Food Programme, 2026a,c).

The funding contraction is sharper here than in the DRC, and it is a contraction in coverage rather than in the value of an individual transfer. WFP reports that it can reach

⁹The impact evaluation is pre-registered with the AEA RCT Registry (Trial #10350); see <https://www.socialscisearch.org/trials/10350>.

¹⁰Under DRC’s administrative structure (Organic Law n° 08/016 of 7 October 2008), provinces sit at the top, followed by territories and communes, then groupements and quartiers, with villages as the next rural tier. Blocs are informal subdivisions below villages used for operational planning and program implementation; they do not correspond to an official administrative level.

around 2 million people a month, down from 4 million the previous winter, and that it requires US\$540 million through January 2027 to sustain its operations (World Food Programme, 2026a). The operational question in Afghanistan is therefore the one this paper prices: when the budget will not stretch to the current caseload at the current package, which features of the package can be changed at least cost to the people receiving it.

This study is embedded in the 2024–2025 round of WFP’s Emergency Food Assistance programme, under Strategic Outcome 1 of its 2018–2025 Country Strategic Plan, and specifically in that programme’s General Food Assistance component: a winter seasonal response paid monthly from November through April. Households enter the caseload in three steps — geographic targeting on food-insecurity indicators, community consultation, and criteria-based selection by sub-district Community Food Assistance Committees, whose lists cooperating partners verify. The cycle reached about 850,000 households nationally.

The transfer value is set from the cost of the food Minimum Expenditure Basket for a reference household of seven, and revised on inflation triggers agreed with the Food Security and Agriculture Cluster and the Cash and Voucher Working Group. The full food basket cost 5,800 AFN per month; funding constraints meant this cycle delivered half rations, 2,900 AFN per household per month over six payments, or 50% of the food basket. Unlike the DRC’s per-member transfer, the Afghan value does not vary with household size.

Unlike the DRC, where cash is delivered in hand and the policy margin under test was who is selected, the Afghan operation has been experimenting with how the transfer arrives. The impact evaluation the study is embedded in randomized the delivery modality, at a different unit in each site. In Kabul’s Police District 12 the unit was the household: 724 households were assigned to cash in hand, 738 to a transfer to an account with Azizi Bank, and 733 to the Hesabpay digital wallet, with all three arms present in every gozar. In the rural site the unit was the village and assignment was stratified on the final distribution point: 89 of the 179 villages received cash and 90 received an in-kind food ration, with no village split across arms. Every sampled household receives assistance: the evaluation has no

untreated arm, excluded on ethical grounds, so what it identifies is the effect of one delivery modality against another rather than of assistance against none.¹¹ That design is the reason the choice experiment’s third modality is site-specific — a bank deposit in Kabul and an in-kind option in the rural site — and it is what allows Section 6.4 to ask whether receiving a modality changes how it is valued.

The transfer used to anchor the Afghan choice tasks is 3,000 AFN, within 3.4% of the 2,900 AFN the programme paid each month, so the willingness-to-pay estimates below are shares of very nearly the transfer respondents were receiving. The reductions the experiment prices span the kind of cut the programme had itself just made in moving to half rations.

Two features of the Afghan setting shape the interpretation of what follows. First, digital and bank-mediated delivery is an actual programmatic option here in a way it is not in rural Tanganyika, so a stated preference over modality in Afghanistan is a preference over alternatives respondents have encountered. Second, respondents are of both sexes, whereas the DRC sample is female by design; this is what lets the paper separate the sex of the recipient from the sex of the person answering.

4 Study Design

4.1 Sample

The DRC sample comprises 3,460 households across 33 villages and 84 blocs in Tanganyika province, eastern DRC. The sample was drawn at the bloc level, with both program beneficiaries and non-beneficiaries of a WFP general food assistance program randomly selected within each bloc. The respondent was the female household head or, if she was unavailable, the primary female caregiver.

¹¹We follow the evaluation report in describing the comparison arm as an in-kind ration — wheat flour, pulses, vegetable oil and salt — though the programme data label it a commodity voucher. For brevity we refer to the rural site as Mazar-i-Sharif, the label the programme data carry and the nearest city; the sampled districts are Almar in Faryab province and Dehdadi and Char Bolak in Balkh, so the site is rural rather than the provincial capital itself.

The survey was administered face-to-face to the female head of household in each respondent household, in the local language, following the enumerator training and piloting protocols described below. Data were collected between September and October 2023 through computer-assisted personal interviews (CAPI). Enumerators completed a one-week training and field piloting, covering survey ethics, standardized questioning, and the administration of the discrete choice experiment. They were also provided with a dedicated manual explaining each of the experiment’s labels and attributes to ensure consistent implementation across interviews.

The study received ethical approval from the University of Milano-Bicocca (#759), a public university affiliated with a principal investigator of the parent evaluation. Respondents provided informed consent prior to the survey, and refusal had no bearing on eligibility for WFP support. Interviews were conducted privately at respondents’ homes, with gender-balanced enumerator teams where possible to protect respondent comfort and privacy. Before the choice tasks, respondents were shown an example choice set and asked comprehension questions; only those who demonstrated understanding of the attributes and choice format proceeded to the DCE module.

The Afghanistan sample comprises 6,486 respondents across two sites: 2,194 in Kabul and 4,292 in Mazar-i-Sharif. Each completed the same six choice tasks over the assistance-package experiment as respondents in the DRC. Transfer levels were 3,000, 2,500, 2,000 and 1,500 AFN, encoding the same 100, 83, 67 and 50% of the full transfer as the DRC design, and the site-specific third modality was a bank deposit in Kabul and a voucher in Mazar.¹²

Respondents are the midline round of the WFP impact evaluation described in Section 3.2, and the sampling frame is that evaluation’s beneficiary roster: enumerators were

¹²At the operational rate the instrument itself used (71.43 AFN per USD, the rate implied by presenting 3,000 AFN as \$42.00), the four levels are \$42.00, \$35.00, \$28.00 and \$21.00. The World Bank’s exchange-rate series (PA.NUS.FCRF) has no Afghanistan observation after 2020, but the instrument’s rate is corroborated externally: the Cash and Voucher Working Group costs the Minimum Expenditure Basket at 71 AFN per USD (UN Treasury, as used by the Afghanistan CVWG, 30 April 2024), within half a percent of it. The evaluation report’s own dollar figures imply a slightly weaker rate again, which is one reason willingness to pay is reported primarily as a share of the full transfer, a metric that requires no conversion.

routed to named households enrolled at baseline and confirmed identity against the roster before proceeding. Fieldwork ran from January 28, 2025 to February 26, 2025 in Kabul and from February 01, 2025 to February 27, 2025 in Mazar-i-Sharif. Interviews were administered face-to-face on tablets, overwhelmingly at the respondent’s home (2,180 of 2,197 in Kabul and 4,021 of 4,308 in Mazar; the remainder at a distribution point or other central location), in the respondent’s own language — predominantly Pashto in Kabul (1,704 interviews, against 486 in Dari) and Dari in Mazar (2,439, with 1,086 in Pashto and 752 in Uzbek). The Kabul site covers 35 neighbourhood clusters within a single police district of Kabul city; the Mazar site covers 179 village clusters across 3 districts served by 8 distribution points.

Of 6,505 households approached, 6,487 consented and 18 declined. The survey was implemented by ARADA (Agency for Rehabilitation Assistance and Development of Afghanistan) on WFP’s behalf. Respondents were told the interview would take 60 to 90 minutes, that participation was voluntary, that they could decline any question or withdraw at any point, and that refusal carried no penalty and no loss of assistance; identifying information was not retained beyond the completion of the research. Ethical oversight was provided by HML IRB under protocol #2706, whose contact details were given to every respondent as part of the consent script.¹³

Enumerators in Afghanistan, as in the DRC, completed a training and piloting programme before fieldwork, covering survey ethics, standardised questioning and the administration of the choice experiment. The two instruments differ in one respect worth stating rather than glossing. The DRC questionnaire walks the respondent through a worked example choice set and asks comprehension questions before the choice tasks begin; the Afghan questionnaire has no equivalent item, presenting the six tasks directly, each accompanied by a visual card and with the block assigned by a random draw inside the form. Comprehension was therefore established in Afghanistan through enumerator training rather than through

¹³The Kabul and Mazar instruments carry identical consent text. The estimation sample is one respondent short of the consenting sample in Mazar — 4,293 consented, 4,292 enter the analysis — because 228 choice rows failed the merge onto the experimental design and were dropped, which removes a single respondent. Kabul’s 2,194 consenting respondents all enter.

a screen inside the instrument.

The Afghan samples differ from the DRC sample in one respect that matters for identification. In the DRC the respondent is the female head or primary female caregiver by design, so every respondent is a woman; in Afghanistan respondents are of both sexes, 42% female in Kabul and 27% in Mazar. Because the DRC sample holds respondent sex fixed, the coefficient on recipient gender cannot there be separated from the sex of the person answering; the Afghan variation is what allows the two to be distinguished, and is one reason to pool the sites at all.

4.2 Conjoint experimental design

We elicit beneficiaries’ preferences over the design and targeting of humanitarian assistance using two discrete choice experiments (DCEs). DCEs build on random utility theory (McFadden, 1974) and the characteristics approach to consumer demand (Lancaster, 1966), under which respondents choose among alternatives defined by varying attribute levels and these choices reveal how much they value each attribute. By varying the attribute levels experimentally and independently of one another, we can use DCEs to identify the marginal effect of each attribute on utility and recover respondents’ marginal valuations of the attributes presented to them (Train, 2009). We apply this framework to two decisions: how assistance is designed, and how would beneficiaries prefer communities committees for targeting are form and take decisions. In both DCEs, respondents are repeatedly presented with two hypothetical alternatives that differ across a small number of randomly varied attributes and choose the one they prefer.

In the first DCE, we apply this framework to the design of humanitarian assistance, recovering the trade-offs beneficiaries are willing to make between program features and, in particular, their willingness to give up transfer value for non-monetary improvements in program design. Concretely, respondents are repeatedly presented with two hypothetical assistance packages that differ across a small number of randomly-varied attributes and

asked to choose the one they would prefer. The first experiment varies the design of the assistance package itself (administered in DRC and Afghanistan); the second, administered only in DRC, varies the design of the committee that selects beneficiaries. Both DCEs use D-optimal designs generated to maximize the determinant of the information matrix subject to attribute-level constraints, and identify all main effects under a multinomial logit framework. Because the objective is to elicit preferences over program features rather than participation decisions, no opt-out option is provided. Respondents are randomly assigned to one block per experiment at the start of the survey, see only the choice tasks within that block, and are asked to compare two profiles per task.

Transfer-attribute experiment.

The first experiment elicits preferences over the design of an unconditional monthly assistance package. The same five attributes appear at all sites: transfer size, frequency, modality, recipient gender, and targeting method. Two of them take site-specific levels. Transfer size ranges over 15,000, 20,000, 25,000 and 30,000 CDF in the DRC (a full transfer of about \$12.82) and over 1,500, 2,000, 2,500 and 3,000 AFN in Afghanistan (about \$42.00). Although these levels differ in nominal terms, they encode the same theoretical trade-offs. The four transfer values correspond, relative to the transfer the household receives under the existing program—which we call the status quo transfer—to no reduction (the full, status quo transfer) and reductions of 17%, 33%, and 50%. Because the transfer values share this percentage scale, we use it as the price metric in the willingness-to-pay analysis. The modality attribute always includes cash-in-hand and mobile money, alongside a site-specific third option: an in-kind food parcel in DRC, a bank deposit in Kabul, and a voucher in Mazar. Although this option varies across sites, the levels encode the same theoretical trade-off, spanning more and less liquid forms of support.

The full attribute-level grid appears in Table 1, and the local-currency levels used in the instruments are reported in Appendix Table A1. Every attribute level was presented to respondents as a graphical representation rather than as text, and the full set is presented

in Appendix A1 as well as a choice set example for each context in Appendix A5. Both instruments use six blocks of six tasks, for 36 distinct choice situations per site. In each task, respondents choose between two packages by answering, “Which assistance package would you prefer?”

Table 1: Attributes and levels — transfer-attribute experiment

Attribute	DRC			Kabul			Mazar-i-Sharif		
	Levels			Levels			Levels		
Transfer size	30,000, 15,000 CDF	25,000, 15,000 CDF	20,000, 15,000 CDF	3,000, 1,500 AFN	2,500, 1,500 AFN	2,000, 1,500 AFN	3,000, 1,500 AFN	2,500, 1,500 AFN	2,000, 1,500 AFN
	(full transfer; 17%, 33% and 50% reductions — identical across sites)								
Frequency	Monthly, bi-monthly			Monthly, bi-monthly			Monthly, bi-monthly		
Modality	Cash in hand, mobile money, in-kind			Cash in hand, mobile money, bank deposit			Cash in hand, mobile money, voucher		
Recipient	Male, female			Male, female			Male, female		
Targeting method	Data-driven, community-based			Data-driven, community-based			Data-driven, community-based		

Notes: The four transfer-size levels are encoded in percentage terms (full transfer; 17%, 33%, and 50% reductions), identically at all three sites. We exploit this encoding throughout the analysis: the mixed-logit price metric is foregone transfer in % of the status quo transfer (Section 5.3), which is what makes the three sites comparable without a currency conversion. The third modality option is site-specific and is the omitted category in estimation, shown in bold; it is the least liquid option available in each setting. Local-currency levels and their USD equivalents are reported in Appendix Table A1.

Targeting-committee experiment.

The second experiment, implemented only in DRC, elicits preferences over the design of a community committee that selects beneficiaries. It was administered to the full DRC sample, immediately after the transfer-attribute experiment and independently of how the respondent answered it, so the committee estimates describe the same population as the transfer esti-

mates rather than a subgroup already sympathetic to community selection. Three attributes vary across profiles: composition (anyone, elected representatives, community leaders), gender balance (mixed, all-female, all-male), and deliberation process (public versus private); the full attribute-level grid appears in Table 2. The instrument uses four blocks of six tasks each (24 distinct choice situations), drawn from the same D-optimal design procedure used for the transfer-attribute experiment. In each task respondents were asked, “*Which committee would you prefer?*” As in the transfer-attribute experiment, every attribute level was presented graphically: the full set of committee-design images appears in Appendix Figure A4, and an example choice task in Appendix Figure A6.

Table 2: Attributes and levels — targeting-committee experiment (DRC)

Attribute	Levels
Composition	Anyone can join, elected representatives, community leaders
Gender	Mixed, all-female, all-male
Process	Public, private

Notes: Six choice tasks per respondent; 4 blocks $\times$ 6 questions; D-optimal design.

Balance and design checks for both experiments are reported in the Appendix.

4.3 Measures

The survey instruments used to collect the choice data also captured a rich battery of household demographic characteristics and outcomes such as food security and coping indicators, and modules on women’s agency and social cohesion. We use these to explore heterogeneity in preferences, restricting the analysis to indicators common across instruments.

Demographic characteristics. From the household roster we record the respondent’s age, educational attainment (coded as an indicator for having at most primary education), and household size and composition. In DRC, the respondent is by design the female head of household or primary female caregiver, but she can still report her husband or male partner

as the head of the household.

Food security and coping. We capture food security with the Food Consumption Score (FCS), which measures the variety and frequency of food groups a household consumed in the seven days before the interview; following WFP thresholds, we classify households with an FCS of 28 or below as having poor food consumption (WFP, 2008). We measure food-related coping with the reduced Coping Strategies Index (rCSI), which captures the frequency of five coping behaviors households adopt when food is short (Maxwell and Caldwell, 2008). Longer-term coping is captured by the Livelihood Coping Strategies Index, an indicator for whether the household resorted to any crisis- or emergency-level strategy. Higher values on each indicate greater food-security stress.

Women’s agency. A decision-making module captures the respondent’s involvement in household decisions (over food, expenditures, and assets) and her autonomy of movement.

Social cohesion and trust. A social-cohesion module captures the relational dimension of vulnerability. From the items common across sites we construct a generalized-trust indicator capturing whether the respondent feels that most people can be trusted.

4.4 Descriptive statistics

Table 3 reports descriptive statistics for the DRC sample of 3,460 households. The DCE respondent is the female head or primary female caregiver by design; the reported household head is male in 61% of households and on average 40 years old. Educational attainment is low: 87% of household heads have at most a primary education. Households are large, averaging 6.5 members.

The sample is highly food insecure and exposed to substantial livelihood stress. 84% of households have poor food consumption ($FCS \leq 28$), and 57% fall in the highest reduced-coping (rCSI) category. Four in five households (80%) resort to crisis- or emergency-level livelihood coping. Generalized trust is low: only 36% of respondents feel that most people can be trusted. Finally, 67% of households reported receiving assistance from WFP.

Table 3: Descriptive statistics — DRC sample

	DRC	
	Mean	SD
Age of household head	39.79	13.98
Male-headed household (= 1)	0.61	0.49
Household head education $\leq$ primary (= 1)	0.87	0.33
Household size	6.51	2.31
Poor food consumption, FCS $\leq$ 28 (= 1)	0.84	0.37
High coping, rCSI $>$ 18 (= 1)	0.57	0.49
Crisis/emergency coping strategy (= 1)	0.80	0.40
Life satisfaction (Anderson index)	0.00	1.00
Generalized trust, most people trusted (= 1)	0.36	0.48
Receiving WFP assistance (= 1)	0.67	0.47

Source: Authors' calculations on the DRC baseline and midline surveys.

Table 4 places the two Afghan samples alongside the DRC one on the characteristics measured at all three sites. Several of the DRC indicators have no Afghan counterpart — the livelihood coping index, the Cantril ladder, generalized trust and prior aid receipt are not in the Afghan instrument — so the common set is narrower than Table 3, and the comparison is confined to it rather than padded with empty cells. Two differences matter for what follows. Respondents are female by design in the DRC but of both sexes in Afghanistan, which is what identifies recipient gender separately from respondent gender. And agency and mobility are measured with different item sets in each country, so the indicators are within-site median splits: a respondent's position relative to their own site is comparable across columns, the level is not.

Table 4: Respondent and household characteristics, by site

	DRC		Kabul		Mazar-i-Sharif	
	Mean	SD	Mean	SD	Mean	SD
Respondent is female	1.00	0.00	0.42	0.49	0.27	0.44
Respondent age (years)	35.13	12.98	40.29	13.66	42.05	13.59
Respondent has low education	0.91	0.29	0.74	0.44	0.91	0.29
Household size	—	—	8.06	2.72	7.66	2.64
Female-headed household	—	—	0.10	0.30	0.10	0.30
Dependency ratio	—	—	0.54	0.19	0.60	0.17
Food insecure (FCS)	0.84	0.37	0.17	0.37	0.32	0.47
High coping-strategy index (rCSI)	0.57	0.49	0.48	0.50	0.24	0.43
Can travel without permission	0.11	0.31	0.23	0.42	0.36	0.48
Agency index above site median	0.49	0.50	0.44	0.50	0.50	0.50
Respondents	3,460		2,197		4,308	

Covariates defined at all three sites. Binary indicators are proportions; age, household size and the dependency ratio are levels. The DRC instrument carries several further measures — the livelihood coping index, the Cantril ladder, generalized trust, and prior aid receipt — that the Afghan instrument does not, and those appear in Table 3 rather than here.

Household size, female headship and the dependency ratio are not carried in the DRC covariate file; the DRC values for the first two are reported in Table 3.

Agency and mobility are measured with different item sets in each country, so only a respondent’s position relative to their own site median is comparable across columns, never the level. The agency indicator is a within-site median split for that reason.

Respondent counts are the surveyed samples. The estimation samples are marginally smaller at the Afghan sites; the counts actually estimated on are reported in Table 5.

5 Framework and Empirical Strategy

5.1 Conceptual framework: transfer value and program design

A policymaker allocating a fixed budget chooses both how much to transfer and how to deliver it. The value dimension is the transfer amount T ; the design dimensions are everything else about the package. Traditionally, this policymaker assumes that a beneficiary’s utility responds only to the monetary value of the transfer, so that

$$U_i = u(y_i + T), \quad (1)$$

where y_i is the non-transfer income of beneficiary i .

We argue that the beneficiary’s true utility also responds to design margins that (1) omits, for instance the delivery characteristics of the transfer or the method used to select recipients. Replacing (1) with the extended specification yields

$$U_i = \underbrace{u(y_i + T)}_{\text{value}} + \underbrace{v_i(\mathbf{a})}_{\text{delivery attributes}} + \underbrace{w_i(p)}_{\text{targeting method}}, \quad (2)$$

where $\mathbf{a}$ stacks the delivery attributes (modality, frequency, recipient gender) and p denotes the targeting method (data-driven versus community-based selection). Equation (2) puts design and value on a single scale: a change in $\mathbf{a}$ or in p now carries a welfare change denominated in units of T , and so can be weighed directly against any change in the transfer amount. These welfare costs and benefits can then enter cost-effectiveness decisions — including, as one application, decisions about how to absorb a reduction in program budgets (Section 6.7).

The choice experiment varies $\mathbf{a}$, p , and T jointly, so we are able to estimate the WTP ratio $-\hat{\beta}/\hat{\beta}_{\text{pct-loss}}$ that recovers the welfare-equivalent transfer foregone for any change in $\mathbf{a}$ or p .

5.2 Preferences

To take (2) to the data, we express it as a random utility model. The utility respondent i obtains from package j in task t is

$$U_{ijt} = x'_{ijt}\beta_i + \varepsilon_{ijt}, \quad (3)$$

where x_{ijt} collects the transfer-size levels and the delivery and targeting attributes $(\mathbf{a}, p)$ of (2), β_i are respondent-specific marginal utilities, and ε_{ijt} is an i.i.d. Respondent i chooses the package that maximizes utility, so the probability of choosing j takes the logit form

$$P_{ijt} = \frac{\exp(x'_{ijt}\beta_i)}{\sum_k \exp(x'_{ikt}\beta_i)}. \quad (4)$$

We estimate (4) in two steps. We first fit a *conditional logit* that constrains tastes to be common across respondents ($\beta_i = \beta$); this serves as a benchmark and as the basis for the Hausman–McFadden test of the independence-of-irrelevant-alternatives (IIA) assumption reported in Appendix Table A3. That test is inconclusive in our data — the estimated difference between the two covariance matrices is not positive definite, which yields a negative statistic and so neither supports nor rejects IIA — so we do not rest anything on it. We therefore estimate a *mixed logit* that lets tastes vary across respondents, $\beta_i \sim f(\beta \mid \theta)$, relaxing IIA by construction and accommodating unobserved preference heterogeneity (Train, 2009). In the mixed logit every attribute, including the three transfer-size levels, enters as a normally distributed random coefficient, while the constant and site fixed effects are held fixed. All attributes are dummy-coded against the reference package, and transfer size enters as three levels (17%, 33%, and 50% below the status quo transfer; the full transfer is the omitted category). We estimate by simulated maximum likelihood with 100 Halton draws and report two specifications: a pooled model that combines DRC, Kabul, and Mazar with site fixed effects, and site-specific models estimated separately for each context. Standard

errors are clustered at the respondent.

5.3 Willingness to pay

We summarize each attribute as a willingness to pay (WTP), the marginal rate of substitution between the attribute and the transfer value. The WTP is the share of the status quo transfer a respondent would forgo to obtain attribute level a , given by

$$\widehat{\text{WTP}}_a = - \frac{\hat{\beta}_a}{\hat{\beta}_{\text{pct-loss}}}. \quad (5)$$

For the WTP estimates we re-specify the mixed logit so that transfer size enters as a single continuous percentage-loss term whose coefficient $\hat{\beta}_{\text{pct-loss}}$ is held *fixed*, while the non-transfer attributes remain random; fixing the price coefficient keeps the denominator of (5) stable and yields well-behaved WTP ratios (Train, 2009). The price metric is the percentage loss from the status-quo transfer in each site. This makes WTP comparable across DRC, Kabul, and Mazar without exchange-rate adjustment. We report WTP as a percentage of the “full transfer” households currently receive (Table 5) and in nominal USD (Table A10). Standard errors come from the delta method and are clustered at the respondent level.

5.4 Heterogeneity

We examine heterogeneity in both experiments. For each characteristic in the common heterogeneity set defined in Section 4.3—food insecurity, coping intensity, education, age, wellbeing, trust, prior aid receipt and agency—we re-estimate the mixed logit on subsamples defined by the indicator. For the transfer-attribute experiment we compare willingness-to-pay profiles; for the targeting-committee experiment, which contains no monetary attribute, we compare utility coefficients within each arm, since a coefficient there is identified only up to that arm’s own error scale. The committee splits add one indicator the transfer splits cannot use: whether the respondent’s bloc was randomized into community-based targeting,

which asks whether experience of a committee shapes preferences over how a committee should be run.

Two limitations of these splits are worth stating in advance. They are neither orthogonal to one another nor randomly assigned—the sole exception being the randomized targeting arm—so the contrasts describe correlations rather than effects. And three of the indicators are missing for part of the sample, so their two arms do not sum to the full sample: generalized trust is missing for 316 respondents and prior aid receipt for 193, and the “no prior aid” arm contains only 133 respondents, which is the thinnest cell we report and should be read with corresponding caution.

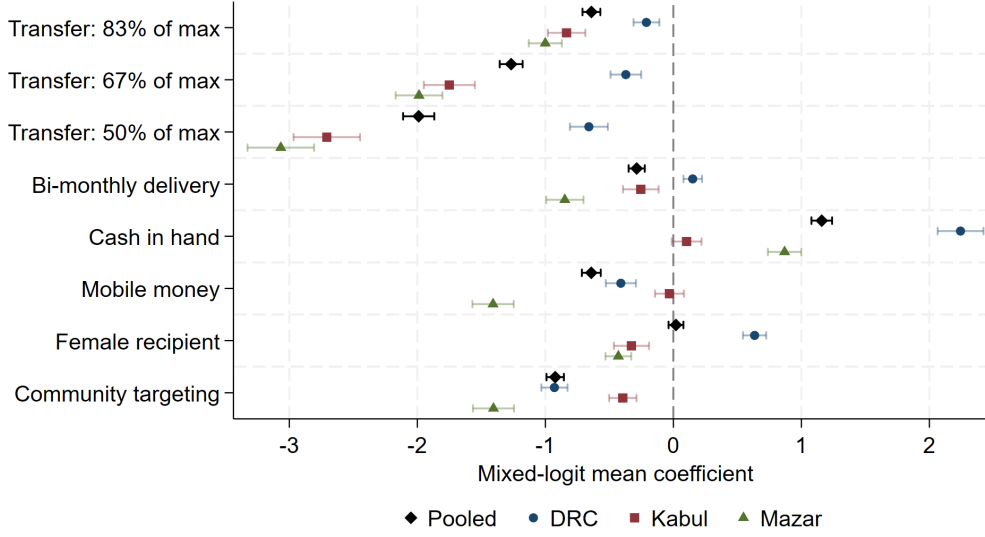
6 Results

6.1 Preferences for transfer attributes

Pooled preferences.

Figure 1 reports the pooled mixed-logit estimates, combining all three sites with site fixed effects. Transfer size is encoded as percentage of the site transfer that households currently receive so coefficients on the size attribute are directly comparable across sites; coefficients on the remaining attributes are relative to the indicated reference category. Positive estimates indicate that an attribute level increases indirect utility relative to the reference, while negative estimates imply welfare losses. Full estimates, including the random-coefficient standard deviations that capture unobserved preference heterogeneity, are reported by site in Appendix Table A5.

Figure 1: Preferences for transfer attributes, pooled and by site



Pooled model carries site-specific alternative-specific constants.
Logit coefficients are identified up to scale, which may differ across sites; see the scale test.
Omitted modality is site-specific: in-kind (DRC), bank (Kabul), voucher (Mazar).

Notes: Mean mixed-logit coefficients with 95% confidence intervals, from the pooled model and from each site estimated separately. The pooled model carries site-specific alternative-specific constants. Reference categories are the full transfer, monthly delivery, the site-specific third modality (in-kind food in the DRC, bank deposit in Kabul, voucher in Mazar), male recipient, and data-driven targeting; modality coefficients are therefore measured against a different good in each site. Logit coefficients are identified only up to scale, which may differ across sites. Standard errors are clustered at the respondent level. Random-coefficient standard deviations are reported in Appendix Table A5.

Transfer size behaves as theory requires. Each successive reduction lowers utility, monotonically and precisely: a 17% reduction by 0.64, a 33% reduction by 1.27, and a 50% reduction by 1.99 (all $p < 0.01$). Per percentage point forgone the implied disutility rises only mildly across the three levels, so the schedule is close to linear over the range respondents were shown, with a modest steepening at the deepest cut. A likelihood-ratio test does not reject the linear restriction ($\chi^2(2) = 1.67$, $p = 0.433$), and both information criteria marginally prefer it (Appendix Table A17). We nonetheless let the size attribute enter as a set of dummies throughout, because the dummy specification nests the linear one and imposes no functional form on a schedule we have no prior about; the cost is two parameters

and the estimates are materially unchanged either way. The random-coefficient standard deviation is significant only at the deepest reduction ($\sigma = 1.69$, $p < 0.01$): households agree about small reductions and diverge sharply about large ones.

Delivery modality carries the largest effects in the design. Cash in hand is strongly preferred to the site-specific third modality ($\hat{\beta} = 1.16$, $p < 0.01$), and in the DRC, where that third modality is in-kind food, the premium is roughly twice as large again ($\hat{\beta} = 2.24$, $p < 0.01$) — more than three times the disutility of a 50% reduction in that site. Mobile money runs the other way: respondents disprefer it relative to the third modality ($\hat{\beta} = -0.64$, $p < 0.01$). The appetite for cash is therefore an appetite for cash in hand, not for digital delivery as such, a distinction that bears directly on any operation weighing a shift to mobile transfers. Both modality coefficients carry large spreads ($\sigma = 1.72$ and 1.83 , both $p < 0.01$), so the aggregate signs conceal substantial disagreement.

The remaining two attributes are where pooling misleads most. Bi-monthly delivery is dispreferred on average ($\hat{\beta} = -0.29$, $p < 0.01$), but the sign is not common across sites: in the DRC bi-monthly delivery is preferred to monthly ($\hat{\beta} = +0.15$, $p < 0.01$), while it is penalised in Kabul (-0.25) and heavily so in Mazar (-0.85). Recipient gender shows the same pattern more starkly. Pooled, designating a woman as recipient is indistinguishable from zero ($\hat{\beta} = 0.02$, n.s.), but that null is an artefact of aggregation rather than a finding: the DRC coefficient is strongly positive ($+0.64$, $p < 0.01$) while Kabul (-0.33) and Mazar (-0.43) are negative and significant. Averaging across these contexts produces a number that describes no site in the study.

Community-based targeting is a substantial penalty rather than a premium: -0.92 pooled and -0.93 in the DRC (both $p < 0.01$), and negative in all three sites. Beneficiaries prefer data-driven selection to selection by a community committee. This deserves stating plainly, because it cuts against a common presumption in humanitarian programming that community involvement in targeting is valued by recipients; in these three settings it is not, though disagreement is wide ($\sigma = 2.01$, $p < 0.01$). Section 6.5 qualifies the result rather than over-

turning it: beneficiaries dislike committee selection as a rule, yet hold clear preferences over how such a committee should be constituted if one is used.

A position effect, common to all three sites.

The alternative-specific constant on the second profile is negative and significant in every site (-0.18 in the DRC, -0.14 in Kabul, -0.19 in Mazar, all $p < 0.01$): holding the attributes fixed, respondents are more likely to choose the profile shown on the left, which takes 51.4% of choices. Effects of this kind are common in conjoint experiments and are usually read as a mix of reading order and satisficing. Two features make it a nuisance here rather than a threat. Attribute levels are balanced across the two positions — no level differs between the left and right profile at even the 10% level (Appendix Table A18) — so the constant absorbs the position effect rather than contaminating the attribute coefficients. And its site interactions are small and insignificant (Table A5), so the bias is common across the three samples rather than something that could masquerade as a cross-site preference difference. We retain the constant in every specification and do not interpret it.

Taken together, beneficiaries’ utility is neither one-dimensional nor uniform. Movements along the design margins produce welfare changes comparable to — and for modality substantially larger than — non-trivial reductions in transfer value, which is what makes the money-metric comparison in Section 6.2 meaningful. Every attribute except the two shallowest transfer reductions carries a significant random-coefficient standard deviation, so the welfare cost of any given design change varies systematically across respondent types, a pattern we exploit in the heterogeneity analysis (Section 6.3).

Site comparisons.

Columns (2)–(4) of Appendix Table A5 estimate the model separately in each site, and they differ enough that the pooled column should be read as a summary rather than as a common preference structure. We test this directly. Re-estimating the pooled model with a full set of site $\times$ attribute interactions and testing whether those interactions are jointly zero rejects equality decisively: $\chi^2(16) = 1219.2$ ($p = 0.001$) for equality across all three sites, with

$\chi^2(8) = 898.6$ for Kabul against the DRC and $\chi^2(8) = 830.3$ for Mazar. The rejection is not driven by one attribute: taken one at a time, every attribute we measure differs significantly across sites, from $\chi^2(2) = 86.4$ for the shallowest transfer reduction to $\chi^2(2) = 486.4$ for cash in hand (Appendix Table A6).

Two cautions belong with that result. First, a mixed-logit coefficient is identified only up to the scale of the error term, so a rejection of equal coefficients is a rejection of equal preferences only if scale is common across sites; Section 6.2 settles this. Second, the omitted modality is a different good in each site — in-kind food in the DRC, a bank deposit in Kabul, a voucher in Mazar — so the modality contrast in particular is measured against a different baseline in each column, and part of the cross-site difference in those coefficients reflects the baseline rather than the alternative.

6.2 Willingness to pay

Table 5 reports willingness to pay — pooled and by site — in % of the status quo transfer foregone, a metric directly comparable across the three sites without currency conversion. Appendix Table A10 translates the same estimates into nominal USD using the site status quo transfer.

Scale versus preference. Table 5 also disposes of the first caution raised in Section 6.1. A mixed-logit coefficient is identified only up to the scale of the error term, so the rejection of equal coefficients across sites is a rejection of equal preferences only if that scale is common. Willingness to pay is immune to the objection by construction: it is the ratio of two coefficients estimated on the same respondents, so a site-specific scale factor multiplies numerator and denominator alike and cancels. Any difference that survives in these columns is a difference in preferences. Differences do survive, and they are not marginal. Designating a woman as recipient is worth 48% of the transfer in the DRC but -7% in Kabul and -8% in Mazar; bi-monthly delivery moves from 14% to -5% and -14% . Both change sign, and no positive rescaling can change a sign. The cash premium changes magnitude on a scale

Table 5: Willingness to pay by site — in % of status quo transfer foregone

	DRC	Kabul	Mazar-i-Sharif
Frequency <i>Ref.: monthly</i>			
Bi-monthly	13.90*** (3.22)	-4.56*** (1.26)	-14.16*** (1.20)
Modality <i>Ref.: site-specific third option</i>			
Cash in hand	169.14*** (19.09)	2.47** (1.07)	14.69*** (1.08)
Mobile money	-30.49*** (5.34)	-0.96 (1.03)	-23.01*** (1.32)
Recipient <i>Ref.: male</i>			
Female	47.78*** (5.69)	-6.78*** (1.24)	-7.80*** (0.87)
Targeting <i>Ref.: data-driven</i>			
Community-based	-68.82*** (8.31)	-7.42*** (0.99)	-22.21*** (1.19)
Observations	41,520	26,328	51,504
No. respondents	3,460	2,194	4,292

WTP is the share of the status quo transfer a respondent would forgo to obtain an attribute, estimated separately for each site. The full transfer is 30,000 CDF (≈ 12.82 USD) in the DRC and 3,000 AFN (≈ 42.00 USD) at both Afghan sites; in each column that transfer is 100%, so the columns are comparable without a currency conversion.

Because WTP is a ratio of two coefficients estimated on the same sample, the scale of the error term cancels: differences across these columns are differences in preferences, not in the precision with which choices are made.

The omitted modality is site-specific — in-kind food in the DRC, a bank deposit in Kabul, a voucher in Mazar-i-Sharif — so the two modality rows are measured against a different baseline good in each column and should be read within a column rather than across.

SEs (delta method) clustered at the respondent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

that rescaling could not plausibly absorb either: 169% of the transfer in the DRC against 2% in Kabul. Community-based targeting is the one attribute on which the three sites agree in direction — -69% , -7% and -22% — and it is dispreferred everywhere. We therefore read the site differences as genuine, and report the by-site columns as primary throughout. A formal scale test in the spirit of Swait and Louviere (1993) would add little: it is a test of a restriction these ratios already reject, and at the size of the pooled sample its rejection is a foregone conclusion. The caveat that does survive is the second one — the omitted modality is a different good in each site — and it applies to the two modality rows, not to the recipient, frequency or targeting contrasts.

These estimates rank the design attributes by the welfare cost of degrading them, and every number below is read from the same macros that populate Table 5, so the text cannot drift from the table.

By a wide margin the most valued feature is the cash modality. Relative to the site-specific third modality — in-kind food in the DRC — beneficiaries value cash in hand at 169% of the full transfer. Recipient gender and the targeting rule come next, and they point in opposite directions: designating a woman as recipient is worth 48% of the transfer, while community-based targeting carries a penalty of -69% . Beneficiaries thus treat who receives the transfer, and how recipients are chosen, as first-order welfare margins rather than administrative details — but they want the transfer directed to a woman and selected by data, not by a committee. The two remaining margins are cheaper to adjust: moving from monthly to bi-monthly delivery is worth 14%, and shifting to mobile money is welfare-reducing relative to the third modality (-30%), consistent with limited mobile-money penetration in this setting. Facing a budget reduction, a planner preserves the most welfare by protecting cash delivery and the recipient and targeting rules, while delivery frequency is the least costly margin to move.

On a willingness to pay above the full transfer.

The cash valuation exceeds 100% of the transfer, which requires comment rather than a

footnote. Taken literally it says a household would give up its entire monthly transfer, and more, to be paid in cash rather than receive an in-kind ration of the same nominal value. Three things about this figure should be stated plainly. First, it is an extrapolation: the design shows respondents reductions of 17, 33, and 50%, so a valuation of 169% lies well beyond the range over which the price term is identified, and the ratio in (5) is linear by construction outside it. Second, it is not an artefact of that linearity. Re-expressing the cash premium against the categorical transfer-size estimates of Table A5 — which impose no functional form — puts it at roughly 3.4 times the disutility of the deepest reduction shown, which is the same order of magnitude. Third, the mechanism is visible in the price coefficient itself: DRC respondents’ utility responds only weakly to the size of the transfer over the range tested, so the denominator of (5) is small and any attribute of substantial value converts into a large share of the transfer.

The economically meaningful reading is therefore ordinal rather than cardinal. In this setting the form of assistance dominates its amount: within the range of reductions a program might plausibly contemplate, beneficiaries would rather absorb a smaller transfer than lose cash delivery. That is a strong claim, and it is the one the data support; the specific figure of 169% should be read as evidence of that ordering, not as a precise monetary equivalence. The by-site estimates reinforce the point by contrast: the same premium is 2% of the transfer in Kabul and 15% in Mazar-i-Sharif — in Kabul almost two orders of magnitude smaller. Two things differ in those columns. The omitted modality is a bank deposit and a voucher rather than in-kind food, so cash is being compared against a much closer substitute; and Afghan respondents trade transfer size off against design far more readily, which prices every attribute lower. The Kabul column is the clearest case: against a bank deposit, cash in hand is worth only 2% and mobile money is statistically indistinguishable from the deposit altogether (−1%). Where the alternative to cash is itself a cash-equivalent instrument, the modality premium largely disappears — which is the substantive counterpart to the DRC finding, not a contradiction of it.

Non-linearity in the welfare cost of cuts.

The disutility of transfer cuts is convex in the depth of the cut: the marginal welfare loss is steeper between the 33 and 50% levels than between the status quo and a 17% cut, though a linear restriction on transfer size is not formally rejected ($\chi^2(2) = 1.67$, $p = 0.433$; Appendix Table A17). We retain the categorical specification, which nests the linear one.

6.3 Heterogeneity in preferences

Tables A7 and A8 report willingness to pay by respondent subgroup along the dimensions described in Section 5.4. Each subgroup is a binary split, the mixed logit is re-estimated separately within each arm on the DRC sample, and the splits are neither orthogonal to one another nor randomly assigned, so the contrasts below are descriptive rather than causal. Three arms are also thin — 133 respondents with no prior aid receipt, 310 in the higher-education arm and 568 in the food-secure arm — and their estimates should be read with that in mind.

Three patterns are worth drawing out.

Deprivation amplifies preferences rather than reordering them. Food-insecure households value cash in hand at 204% of the transfer against 98% for the food-secure, penalise community-based targeting at -88% against -28% , and value a female recipient at 58% against 24%. The ranking of attributes is identical in the two arms; what changes is the intensity. The same gradient appears for low wellbeing (274% versus 149% for cash) and for prior aid receipt (164% versus 31%). This is what one would expect if the marginal utility of the transfer is higher for the more deprived: the same design change buys proportionally more welfare where the baseline is worse.

The dislike of mobile money is concentrated, and it tracks capacity to use it. Mobile money is dispreferred on average, but the sign is not uniform. Households whose head has more than primary education prefer mobile money, at $+23\%$ of the transfer, while those with low education disprefer it at -36% . This is the only significant sign reversal in either

table, and it matters operationally: it suggests resistance to digital delivery is not a fixed preference over money’s form but a function of who can convert a mobile balance into goods. A shift to mobile transfers would be close to welfare-neutral, or better, for the more educated households, and costly for exactly those the program most intends to reach.

The targeting result is robust; the cash result is not fragile either. Community-based targeting attracts a significant penalty in every one of the eighteen subgroup arms, ranging from -27% to -157% . There is no subgroup in the DRC sample that prefers committee selection to data-driven selection, which is a stronger statement than the pooled coefficient alone supports. Cash in hand is likewise valued above the full transfer in sixteen of the eighteen arms. The two exceptions are the food-secure (98%, marginally below the full transfer) and households that had not previously received assistance (31%) — and the latter rests on 133 respondents and is the least precisely estimated cell in either table, so it should not carry much weight. Generalized trust cuts the same way: more trusting households value cash at 296% against 134% for the less trusting.

These splits are exploratory, and the estimates in Tables A7 and A8 are DRC-only.

Does the DRC pattern travel? Because the paper’s central claim is that valuations differ across sites, a heterogeneity pattern established in one site cannot be reported as though it were general. Table A9 therefore re-estimates the subgroup models separately within each site, over the 7 characteristics the three instruments share. The answer is that one finding travels, and the rest do not.

The exception is targeting, and it is emphatic. Community-based targeting carries a significant penalty in *every* cell of the by-site table — forty subgroup-by-site combinations, in three countries, without one exception. Nothing else in the paper is this robust, and it is the finding a programme designer should weigh most heavily.

The cash premium is the opposite case: stable within a site, and separated across sites by an order of magnitude in every single subgroup. Cash in hand is worth between 92% and 209% of the transfer in each DRC arm, between -0.4% and 5% in each Kabul arm,

and between 10% and 17% in each Mazar arm. No subgroup in Afghanistan approaches any subgroup in the DRC. Whatever drives that gap, it is not the composition of the samples on any characteristic we measure — which is what the omitted modality being a bank deposit or a voucher, rather than a sack of food, would predict.

Two DRC results turn out to be local. Food insecurity roughly doubles the DRC cash premium, from 98% to 204%; in Mazar it *lowers* it, from 17% to 10%, and in Kabul it does nothing. And the education reversal in mobile money reported above — the only sign reversal in the DRC tables — does not replicate: in Mazar both education groups dislike mobile money, the less-educated more strongly. A reader given the DRC panel alone would have drawn a general conclusion from a local one in both cases.

Who dislikes a female recipient. The most consequential result in the by-site table concerns recipient gender, because it revises the cross-country contrast of Section 6.1 rather than qualifying it. In the DRC, where every respondent is a woman, a female recipient is valued in all fourteen subgroup arms, from 24% to 70% of the transfer. In Afghanistan the sign is generally negative — but not uniformly, and the exceptions are informative. Afghan women do not share the aversion: female respondents in Kabul value a female recipient at +5.8% while male respondents there place it at −16.2%, and the same split appears in Mazar. Mobility points the same way. Kabul respondents who can travel without asking permission value a female recipient at +10.5%; those who cannot place it at −12.2%.

Two independent indicators — who answered, and whether women in the household move freely — therefore locate the aversion among men and among households where women’s movement is constrained, rather than in Afghanistan as such. This matters for how Section 6.1’s reversal should be read. The share of female respondents is 100% in the DRC, 42% in Kabul and 27% in Mazar, which is the same order as the site coefficients: a substantial part of what looks like a difference between countries is a difference in who was asked. Not all of it, though. Comparing women with women, the DRC figure of 48% still far exceeds Kabul’s 6%, so a genuine contextual difference survives the decomposition. The composition

effect explains the change of sign; it does not explain the distance.

Neither the respondent’s sex nor the household’s mobility is randomly assigned, so these are decompositions rather than identified effects: who answers a survey is a household decision, and may itself reflect the norms that shape the preference. They are nonetheless enough to show that “Afghanistan disprefers female recipients” is the wrong summary of the Afghan estimates.

6.4 Stated versus revealed preferences

As mentioned above, the DRC sample comes from a cluster-randomized trial in which villages were assigned to either a new community-based targeting method or WFP’s traditional data-driven approach (PMT+). The random assignment lets us test whether stated preferences are consistent with the targeting modality households actually experienced. Table 6 re-estimates the willingness-to-pay model separately within each arm.

This is the sharpest available test of the targeting result, and it survives. Community-based targeting is dispreferred in both arms: by -73.7% of the full transfer among households assigned to data-driven selection, and by -62.2% among those assigned to community-based selection (both $p < 0.01$). Direct experience of the mechanism leaves the penalty large, negative, and precisely estimated. The gap between the two arms is not itself distinguishable from zero ($z = 0.71$), so we do not read the smaller penalty in the community-based arm as evidence that exposure softens the objection. Whatever drives it, it is not unfamiliarity.

No single valuation differs across the arms. The largest gap anywhere in the profile is on a female recipient, 38.7% against 54.3% , and even that is $z = 1.42$; the cash premium (172.7% and 166.0%), the aversion to mobile money (-35.3% and -26.2%) and the value of bi-monthly delivery (16.5% and 10.4%) are closer still. A joint test across all five attributes is a different matter: because the arms are independent samples, their difference can be tested directly, and equality of the whole profile is rejected at the margin, $\chi^2(5) = 11.40$ ($p = 0.04$). The honest reading is that exposure does not move any particular valuation by a detectable

amount, but that the five small differences, taken together, are more than sampling noise comfortably explains.

Table 6: Preferences by treatment arm — WTP (% of full transfer)

Site	DRC	
	CBT vs. data-driven targeting	
	Data-driven	Community-based
RCT		
Bi-monthly frequency	16.47*** (4.75)	10.44** (4.15)
Cash in hand	172.66*** (27.08)	166.01*** (26.26)
Mobile money	-35.29*** (7.92)	-26.23*** (6.82)
Recipient: female	38.67*** (6.76)	54.32*** (8.73)
Targeting: community-based	-73.69*** (12.05)	-62.24*** (10.71)
Observations	21,036	20,484

WTP in % of the full transfer a respondent would forgo to obtain an attribute. Columns split the DRC sample by the village-level randomization into the data-driven (PMT+) and community-based targeting arms. SEs (delta method) clustered at the respondent. The arms are independent samples, so their difference is tested directly: a joint Wald test of equality across all five attributes gives $\chi^2(5) = 11.40$, $p = 0.04$, and no single attribute differs by more than $|z| = 1.42$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The Afghan evaluation supplies a second, independent version of the same test on a different margin. It randomized the delivery modality rather than the targeting method: households in Kabul were assigned to cash in hand, an Azizi Bank transfer or the Hesabpay digital wallet, and households in Mazar-i-Sharif to cash or a commodity voucher. If stated preferences over modality were an artefact of what a household happens to be receiving — if people simply endorse the instrument they know — these arms are where it would show.

Table 7 re-estimates willingness to pay separately within each of the five arms.¹⁴

¹⁴The two sites randomized at different units, which bears on the standard errors in the rural columns. Kabul assigned modality household by household, so clustering at the respondent matches the assignment. The rural site assigned whole villages — 89 to cash and 90 to the in-kind ration — so the rural standard

It does not show. In Kabul, the value placed on cash in hand is 4%, 0% and 2% of the transfer among households assigned to cash, to the bank transfer and to the wallet respectively — small everywhere, and with no ordering by what the household receives. Mobile money is likewise close to zero in all three (0%, −4% and 1%). In Mazar the contrast is sharper and runs the wrong way for an adaptation story: households randomized to the commodity voucher value cash at 17% of the transfer, slightly more than the 13% among households already receiving cash, and they disprefer mobile money by a similar margin (−24% against −20%). Receiving a substitute for cash does not reconcile people to it. None of these differences is significant: the largest anywhere in Afghanistan is $|z| = 1.93$, and the joint tests of profile equality do not reject in Kabul ($\chi^2(10) = 12.99$, $p = 0.22$) and only marginally so in the rural site ($\chi^2(5) = 9.91$, $p = 0.08$).

Taken with the DRC result, the evidence rests on two randomizations, in two countries, on two different design margins, and it is consistent rather than uniform. On no margin, in no site, does the arm a household was assigned to change what it will give up for any single attribute. The joint tests are less tidy: Kabul is comfortably consistent with equality, the rural site is borderline, and the DRC rejects at the margin. What the evidence supports, then, is the weaker but still useful claim — that exposure does not reorganize preferences in the way the standard objection to eliciting them supposes. Beneficiaries do not simply ratify the practice they know: in the DRC they disprefer community-based targeting whether or not they have lived under it, and in Afghanistan they value cash no more highly for having received it.

6.5 Preferences for the targeting-committee experiment (DRC)

Appendix Table A11 reports preferences over the three attributes of targeting-committee design: composition, gender, and deliberation process.

Across all three attributes, beneficiaries favor committees with member that are elected errors, also clustered at the respondent, are understated relative to a village-clustered alternative. The finding in those columns is a null, which wider standard errors would only reinforce.

Table 7: Preferences by randomized modality arm, Afghanistan — WTP (% of full transfer)

	Kabul			Mazar-i-Sharif	
	Cash in hand	Azizi Bank	Hesab-pay	Cash	Commodity voucher
Bi-monthly frequency	-1.02 (2.12)	-3.99* (2.12)	-7.01*** (2.27)	-14.64*** (1.57)	-14.55*** (1.84)
Cash in hand	3.99** (1.83)	0.45 (1.83)	2.34 (1.90)	13.46*** (1.44)	16.90*** (1.68)
Mobile money	0.18 (1.77)	-3.88** (1.70)	1.10 (1.80)	-20.42*** (1.71)	-24.37*** (1.97)
Recipient: female	-4.55** (2.00)	-7.42*** (2.14)	-8.05*** (2.27)	-8.91*** (1.26)	-6.50*** (1.21)
Targeting: community-based	-9.20*** (1.81)	-7.60*** (1.61)	-5.22*** (1.67)	-24.14*** (1.78)	-21.54*** (1.79)
Observations	8,676	8,856	8,772	25,776	25,716

WTP in % of the full transfer a respondent would forgo to obtain an attribute, estimated separately within each randomized arm of the Afghan impact evaluation. The evaluation randomized the delivery modality actually received: cash in hand, an Azizi Bank transfer, or the Hesabpay digital wallet in Kabul; cash or a commodity voucher in Mazar-i-Sharif. The omitted modality in the choice experiment is a bank deposit in Kabul and a voucher in Mazar, so cash and mobile-money WTP are measured against the site's own third option and are not comparable across the two sites. SEs (delta method) clustered at the respondent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

by them, that are gender-balanced, and follow a more transparent process. On composition, respondents prefer a committee of elected representatives to one open to anyone ($\hat{\beta} = 0.25$, $p < 0.01$) and disprefer one staffed by community leaders (-0.13 , $p < 0.01$), indicating a demand for accountable selection over both unstructured and incumbent-led arrangements. On gender, the mixed-gender committee is the most preferred: an all-male committee is strongly dispreferred (-0.76 , $p < 0.01$) and an all-female committee mildly so (-0.27 , $p < 0.01$). The sharpest preference is over process: private deliberation is strongly dispreferred relative to public deliberation (-0.96 , $p < 0.01$), the largest committee coefficient in the table. The substantial random-coefficient standard deviations show these procedural preferences are widely held but not uniform. The pattern is telling, and it sits oddly beside the targeting result in Section 6.1: the same respondents who disprefer community-based selection place a first-order value on the procedural legitimacy of how a committee reaches its decisions—elected membership, gender balance, and public deliberation—rather than treating committee design as an administrative detail. What they object to appears to be unconstrained local discretion over who is included, not community participation as such. The distinction becomes practically important whenever a committee must exclude more applicants than before, since it is precisely then that the procedural constraints bind.

Who holds these procedural preferences. Appendix Tables A12 and A13 re-estimate the committee model within each subgroup arm. The headline is how little moves. Private deliberation is disliked in all twenty arms, from -0.72 to -1.54 , and never approaches zero; all-male committees are dispreferred in every arm; and elected composition is preferred in all but one. Whatever drives the demand for procedural legitimacy, it is not confined to a particular kind of household.

Three contrasts are worth noting. The single exception on composition is wellbeing: respondents low on the Cantril ladder are indifferent between an elected committee and one open to anyone (-0.10 , n.s.) while everyone else prefers election ($+0.34$, $p < 0.01$), as though the expectation that a procedure will work for you is itself a function of how your life is going.

Generalized trust runs the other way from the obvious prediction: respondents who think most people can be trusted object more strongly to private deliberation (-1.37) than those who do not (-0.81), so the demand for openness is not a substitute for trust but something closer to a complement. And the randomized targeting arm barely matters: households whose bloc was assigned to community-based targeting value elected membership ($+0.31$) and dislike private deliberation (-0.99) at almost exactly the rates of households that were not ($+0.19$ and -0.93). Direct experience of a committee does not reshape preferences about how committees should work, which is the same conclusion the exposure tests of Section 6.4 reach on two other margins.

6.6 Predictors of preferences

The attribute that carries the most welfare is delivery modality, and the ordering within it is essentially an ordering by liquidity: cash in hand is spendable immediately and anywhere, mobile money once it is cashed out or where it is accepted, and the site-specific third option least of all. This subsection asks which households trade other features of the package away in order to obtain the more liquid one.

Each choice occasion is reduced to a single contrast — did the respondent take the more liquid of the two packages offered? — with the differences between them in transfer value, frequency, recipient and targeting entering as controls, so the household coefficients describe the willingness to give those features up for liquidity rather than a raw taste for cash. The design never offers the same modality twice within a task, so every occasion carries a contrast. The estimation sample is 43,020 occasions from 7,170 respondents: the model is fitted on the DRC and Mazar samples, Kabul being excluded because its third modality is a bank deposit and cannot be ranked for liquidity against cash and mobile money, and a further 582 of the 7,752 eligible respondents drop out through listwise deletion on the covariates — overwhelmingly the 485 DRC respondents for whom freedom of movement is not recorded. That loss is concentrated in one site and one variable, so the coefficient on

freedom of movement in particular describes only the households for whom it was measured. Table A16 reports the estimates.

The result is mostly a null, and it is worth reporting as one. Food insecurity, coping intensity, respondent age, education and freedom of movement are all unrelated to the choice of a more liquid package. The strong heterogeneity documented in Section 6.3 is heterogeneity in how much attributes are valued, not in which package a household picks when liquidity is on the table: the preference for liquidity is close to universal in this sample, and observable household characteristics predict very little of the residual variation (pseudo- $R^2 = 0.073$).

The one characteristic that does predict it does not replicate across contexts. Women with greater decision-making agency are more likely to take the liquid package in the DRC and Mazar sample (+0.115, $p < 0.01$), and less likely in Kabul (-0.132 , $p < 0.01$). We do not think this should be over-interpreted. The agency index is built from different instruments in each site, so only the within-site position is comparable and the two coefficients are not measuring the same underlying quantity; the sign reversal may reflect the instruments as much as the households. We report it because it is the only systematic predictor we find, and because it is a further instance of the pattern that runs through these results: what looks like a household-level regularity in one setting does not survive transport to another.

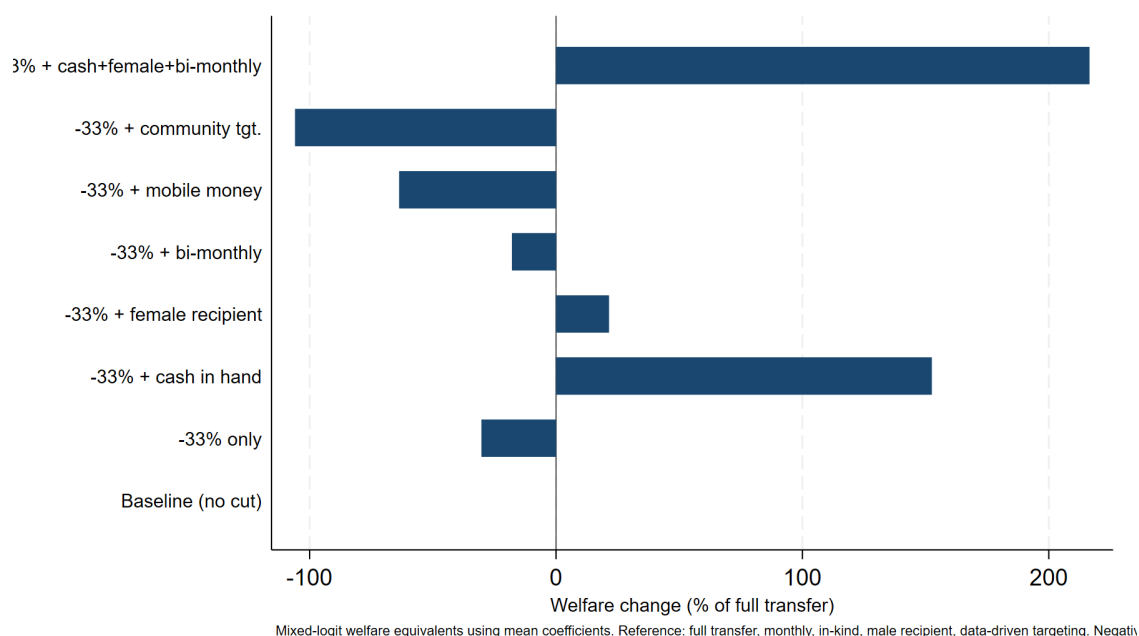
A caveat on the remaining significant coefficient. Respondent sex enters column (1) negatively, but every DRC respondent is female, so with a site fixed effect that coefficient is identified within Mazar alone. It is an Afghan result, not a DRC one, and we do not draw on it further.

6.7 Policy simulation under aid cuts

A central motivation of our design is to recover the welfare implications of plausible aid-cut paths and to identify which design adjustments preserve beneficiary welfare. The transfer-size attribute encodes cuts directly (17, 33, and 50% below the full transfer), while the remaining attributes describe the design margins over which a program can be redesigned.

We use the mixed-logit estimates to simulate the welfare consequences of reduction-and-redesign packages relative to a reference package in which every attribute takes its omitted level: the full transfer, monthly delivery, in-kind food, a male recipient, and data-driven targeting. This reference is the estimation baseline rather than a description of the current program. The transfer amount and the delivery frequency do correspond to what DRC households currently receive, but the modality does not — cash in hand is the prevailing modality in this operation, and in-kind food is the omitted category only because some level must be omitted for the model to be identified. The simulation should therefore be read as pricing movements away from a common reference point, not as costing departures from the status quo. For each scenario we report a welfare-equivalent change, in percent of the full transfer: we translate the difference in indirect utility between the scenario and the baseline into transfer-value units using the average marginal utility of cash, so a negative value is a welfare loss and a positive value means the redesigned package is preferred to the reference.

Figure 2: Welfare-equivalent change under a 33% cut paired with each design switch



Notes: Welfare-equivalent change (% of the full transfer) relative to the reference package, from pairing a 33% transfer cut with each compensating design switch. Computed from the mixed-logit mean coefficients.

Figure 2 traces the welfare consequences of a 33% reduction paired with each compensating design switch. Moving delivery from in-kind to cash in hand is by far the most powerful lever, and it does not merely offset the reduction: the package is worth +152.5% of the full transfer relative to the reference, against -30.2% for the reduction alone. Designating a woman as recipient recovers a smaller but material share (+21.5% net of the reduction). The other switches move the wrong way. Bi-monthly delivery is close to neutral once paired with the reduction (-17.9%), mobile money deepens the loss (-63.6%), and community-based targeting deepens it sharply (-105.9%) — so a program that responded to a smaller budget by devolving selection to community committees would roughly triple the welfare cost of the reduction itself. Combining the preferred switches recovers far more than the loss (+216.5%), and does so even under a 50% reduction (+193.0%), whose unaided cost is -53.7% . Appendix Table A15 decomposes these into the share of the 33% loss each switch recovers on its own.

Two cautions apply to reading these magnitudes. They inherit the extrapolation discussed in Section 6.2 — the cash figure in particular rests on a small DRC price coefficient and lies outside the range of reductions respondents were shown — and they are computed at the mean coefficients, so they describe the average household rather than any particular one. The ordering of the levers is the robust content here; the point estimates are best read as showing that modality and targeting are the margins that dominate, not as precise compensation ratios. Appendix Table A14 reports all scenarios in detail.

7 Discussion

Three results from this design bear on how assistance is written, and each cuts against a common working assumption.

The form of assistance can matter more than its amount. Within the range of transfer values we tested, delivery modality carries larger welfare consequences than the size of the

transfer itself. In the DRC beneficiaries would forgo more than a full monthly transfer to be paid in cash rather than receive an in-kind ration of equivalent nominal value. We are deliberate in Section 6.2 about how much weight that specific ratio can bear — it extrapolates beyond the reductions respondents were shown, and it is driven by a price coefficient that is small in the DRC. But the ordering it implies is robust to how transfer size is specified and holds in every subgroup we examine. For an operation deciding between a smaller cash transfer and a larger in-kind one, this is the first-order consideration, and it is not what a budget line that prices only the ration would suggest.

Beneficiaries do not want to be selected by their neighbours. Community-based targeting is dispreferred relative to data-driven selection in the pooled model, in all three sites, and in all eighteen subgroup arms we estimate. This is the result most at odds with prevailing practice, where community involvement in beneficiary selection is often treated as both participatory and legitimising. Two features of our design make it hard to dismiss. First, the DRC evaluation randomised communities into community-based targeting, and households that actually experienced it still disprefer it (−62% of the transfer) almost as strongly as those that did not (−74%), a difference that is not statistically distinguishable ($z = 0.71$). The penalty is therefore not unfamiliarity, and exposure does not reconcile beneficiaries to the mechanism. Second, the same respondents hold detailed and coherent preferences over how a targeting committee should be built if one is used — they favour elected members over community leaders, gender-mixed committees over single-sex ones, and public deliberation over private (−0.96, $p < 0.01$, relative to public).

We read these together as an objection to discretion rather than to participation. What beneficiaries appear to dislike is a selection process in which local actors exercise judgement over who is included, with the scope for favouritism and social pressure that implies; what they endorse are the features that constrain that discretion — election, gender balance, and public reasoning. A programme that wants community legitimacy without the welfare cost should therefore invest in the procedural constraints rather than in community involvement

as such.

Preferences do not travel, and pooling can manufacture null results. We reject equality of attribute valuations across the three sites for every attribute we measure. The clearest illustration is recipient gender: pooled, designating a woman as recipient is a precise zero, which would ordinarily be reported as “no effect”. It is not a null. The DRC coefficient is strongly positive and both Afghan coefficients are negative and significant, and the pooled estimate simply averages them away. Delivery frequency behaves the same way, preferred in the DRC and penalised in both Afghan sites. Multi-country evidence on assistance design that reports pooled coefficients may be reporting artefacts of aggregation, and the practical implication is that these parameters need to be measured where a programme operates rather than imported.

That said, disaggregation should not stop at the border. Splitting the recipient-gender result by site is an improvement on pooling, but splitting it further shows that the site contrast is itself partly an aggregate: the Afghan aversion belongs to male respondents and to households where women’s mobility is restricted, not to the Afghan samples as wholes, and the DRC sample is female by construction. A site is a container for people, and reporting a site mean is only the second-best answer to the question of whose preference is being measured. The general lesson is the same one twice over — an average describes a population only if the population is homogeneous in the respect that matters, and neither a pooled sample nor a national one need be.

Implications for digital delivery. Mobile money is dispreferred on average, but that average conceals the only sign reversal in our heterogeneity analysis: households whose head has more than primary education prefer mobile money, while those with low education disprefer it substantially. Resistance to digital transfers looks less like a preference over the form of money than a function of who can convert a mobile balance into goods. A shift to mobile delivery would fall most heavily on precisely the households a programme is most concerned to reach, which argues for sequencing such transitions against agent-network

coverage and financial literacy rather than rolling them out uniformly.

Limitations. Four are worth stating. These are stated preferences over hypothetical packages; the stated-versus-revealed comparison in Section 6.4 now covers two randomised margins in two countries — targeting in the DRC, delivery modality in Afghanistan — and neither moves stated preferences; it remains a check on those two attributes rather than on the instrument as a whole. The omitted modality differs across sites — in-kind food in the DRC, a bank deposit in Kabul, a voucher in Mazar — so the modality contrasts are not measured against a common good, and cross-site comparison of those coefficients in particular should be read with care. Mixed-logit coefficients are identified only up to the scale of the error term. That objection is answered for the contrasts we lean on, since the willingness-to-pay estimates of Table 5 are ratios in which a site-specific scale cancels, and they differ across sites by an order of magnitude and in sign; but it does apply to the pooled coefficient vector in column (1), which is a further reason to read the by-site columns as primary. Finally, the heterogeneity analysis is DRC-only and its splits are neither orthogonal nor randomly assigned, so those contrasts describe correlations rather than effects.

8 Conclusion

We ran a conjoint experiment with food-assistance beneficiaries in the DRC and two Afghan cities, varying the delivery modality, frequency, intra-household recipient, and targeting method of assistance alongside the value of the transfer, so that program design and transfer size are priced on a common scale.

Beneficiaries hold strong, well-ordered, and context-specific preferences over how they are assisted, not merely how much they receive. Modality dominates: cash in hand is valued far above the in-kind alternative, while mobile money is dispreferred by exactly those households least equipped to use it. Community-based targeting carries a substantial penalty in every site and every subgroup, and randomised exposure to it does not soften that judgement,

even though the same respondents have clear views on how a targeting committee should be constituted. And valuations differ across the three sites for every attribute we measure, to the point where pooled estimates can report a null that describes no site in the study.

For programmes facing constrained budgets, the practical content of these results is that design margins are not free and are not interchangeable. Section 6.7 prices them directly: some redesigns offset a meaningful share of the welfare cost of a smaller transfer, and others compound it. Deciding which transfer to cut, and how to deliver what remains, are the same decision, and beneficiaries’ own valuations are measurable inputs to it.

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Appendix

Table A1: Transfer levels by basis and local currency

Site	Basis	Level	Local currency	Nominal USD	PPP USD
DRC (Kalemie, Tanganyika) — Sept–Oct 2023					
	Per capita	Full	30,000 CDF	12.82	32.31
		17% reduction	25,000 CDF	10.68	26.92
		33% reduction	20,000 CDF	8.55	21.54
		50% reduction	15,000 CDF	6.41	16.15
Afghanistan (Kabul and Mazar-i-Sharif) — Jan–Feb 2025					
	Per household	Full	3,000 AFN	42.00	—
		17% reduction	2,500 AFN	35.00	—
		33% reduction	2,000 AFN	28.00	—
		50% reduction	1,500 AFN	21.00	—

Notes: The four transfer-size levels are exploited as the price metric in the WTP analysis (Section 5.3) in percentage terms (full transfer; 17%, 33%, and 50% reductions), which is identical across sites and requires no currency conversion. The DRC attribute is denominated per household member; the Afghan attribute is per household, so the two local-currency columns are not directly comparable in level. DRC nominal USD uses the 2023 official exchange rate of 2,340 CDF per USD (World Development Indicators, series `PA.NUS.FCRF`); DRC PPP values use the purchasing-power-parity conversion factor for household final consumption of 928.60 CDF per international dollar (series `PA.NUS.PRVT.PP`, 2021, the latest published for the DRC). Afghan nominal USD uses 71.43 AFN per USD, the operational rate implied by the survey instrument, which presented the full transfer of 3,000 AFN as \$42. No World Bank exchange rate or PPP conversion factor is published for Afghanistan after 2020, so no independent official rate and no PPP column are available for the Afghan survey period.

Every attribute level was shown to respondents as a graphical representation rather than as text. Figures A1 through A4 display the full set of attribute-level images for each context, and Figure A5 shows an example choice task for each context.

Figure A1: Attribute levels presented to respondents — DRC

Forfait d'assistance				
La valeur de l'assistance:	 30,000 CDF ppm	 25,000 CDF ppm	 20,000 CDF ppm	 15,000 CDF ppm
Comment le recevoir:	 Argent liquide en main	 L'argent mobile (sur le telephone)	 Transfert en nature (valeur équivalente)	
À quelle fréquence:	 1 Mensuelle	 2 mois (la quantité de deux mois)		
Qui le reçoit:	 Femme	 Homme		
Comment sélectionner bénéficiaires:	 Basé sur des données/enquêtes		 Processus d'autosélection communautaire	

Figure A2: Attribute levels presented to respondents — Afghanistan (Kabul)

Assistance package				
Value of assistance	 3,000 AFN	 2,500 AFN	 2,000 AFN	 1,500 AFN
What form is receive	 Cash in hand	 Money	 Bank	
How often	1  Monthly	2  (the amount of two months)		
Who receives it	 Man	 Man		
How beneficiaries are selected	 Using a survey	 Community committee		

Figure A3: Attribute levels presented to respondents — Afghanistan (Mazar)

Assistance package				
Value of assistance	 3,000 AFN	 2,500 AFN	 2,000 AFN	 1,500 AFN
What form is receive	 Cash in hand	 Money	 Voucher	
How often	 Monthly	 2 (the amount of two months)		
Who receives it	 Man	 Man		
How beneficiaries are selected	 Using a survey	 Community committee		

Figure A4: Attribute levels presented to respondents — targeting committee

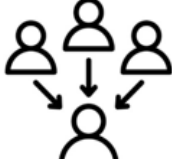
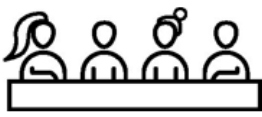
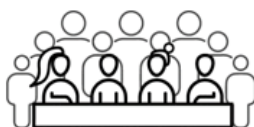
Composition du comité communautaire			
Membres du comité:	 Chef de communauté	 Élu par la communauté	 N'importe qui
Sexe des membres:	 Toutes féminines	 Tous masculins	 Genre mixte
Processus de sélection	 En privé (sans observation de la communauté)	 En public	

Figure A5: Example choice task — transfer-attribute experiment

Forfait d'assistance	A	B
La valeur de l'assistance:	 20,000 CDF ppm	 15,000 CDF ppm
Comment le recevoir:	 Argent liquide en main	 L'argent mobile (sur le telephone)
À quelle fréquence:	 2 mois (la quantité de deux mois)	 Mensuelle
Qui le reçoit:	 Femme	 Homme
Comment sélectionner bénéficiaires:	 Basé sur des données/enquêtes	 Processus d'autosélection communautaire

(a) DRC

Assistance package	A	B
Value of assistance	 2,000 AFN	 1,500 AFN
What form is receive	 Cash in hand	 Mobile money
How often	 2 (the amount of two months)	 Monthly
Who receives it	 Woman	 Man
How beneficiaries are selected	 Using a survey	 Community committee

(b) Afghanistan (Kabul)

Assistance package	A	B
Value of assistance	 1,500 AFN	 3,000 AFN
What form is receive	 Mobile money	 Voucher
How often	 Monthly	 Once every two month (the amount of two months)
Who receives it	 Woman	 Man
How beneficiaries are selected	 Community committee	 Using a survey

(c) Afghanistan (Mazar)

Figure A6: Example choice task — targeting-committee experiment (DRC)

Composition du comité communautaire	A	B
Membres du comité:	 Élu par la communauté	 Chef de communauté
Sexe des membres:	 Tous masculins	 Genre mixte
Processus de sélection	 En privé (sans observation de la communauté)	 En public

Table A2 reports the standard design-efficiency metrics returned by the D-optimal optimization for each conjoint experiment. Lower values of the D-error (determinant of the asymptotic variance-covariance matrix of the parameter estimates) and the A-error (its trace) indicate a more informative design.

Table A2: Design-efficiency metrics by conjoint experiment

Metric	Transfer-attribute	Targeting-committee
D-error	0.1766	0.2590
A-error	0.2226	0.3008
B-estimate	—	100
S-estimate	—	0

Notes: D-error and A-error are the determinant and trace of the asymptotic variance–covariance matrix of the parameter estimates implied by the D-optimal design; lower values denote greater information content. B-estimate and S-estimate are Bayesian-evaluation diagnostics returned for the targeting-committee design only.

Table A3: Test of independence of irrelevant alternatives — preferences (pooled)

	Full	Deducted	Difference	S.E.
Transfer: 17% less	-0.14	-0.17	0.03	0.00
Transfer: 33% less	-0.17	-0.20	0.03	0.00
Transfer: 50% less	-0.32	-0.34	0.02	0.01
Frequency: bi-monthly	0.07	0.06	0.01	0.00
Cash in hand	0.94	0.92	0.03	0.01
Mobile money	-0.14	-0.13	-0.00	0.00
Recipient: female	0.28	0.27	0.01	0.00
Targeting: community-based	-0.43			
Constant				
<i>Observations</i>	41,520	41,520		
<i>No. respondents</i>	.	.		
<i>Log-likelihood</i>	-12255.01	-12638.34		
Hausman test:	$\chi^2(8) = -65.68, p = 1.000$			

Table A4: IIA / Hausman test — targeting committee

	Full	Deducted	Difference	S.E.
Composition: elected	0.11	0.11	0.00	0.00
Composition: community leaders	-0.07	-0.06	-0.01	0.01
Gender: all-female	-0.13	-0.13	0.00	0.00
Gender: all-male	-0.41	-0.39	-0.02	0.01
Process: private	-0.51			
<i>Observations</i>	41,520	41,520		
<i>No. respondents</i>	.	.		
<i>Log-likelihood</i>	-13454.62	-14090.75		
Hausman test:	$\chi^2(5) = -40.59, p = 1.000$			

Table A5: Preferences for assistance attributes — pooled and by site

	(1) Pooled	(2) DRC	(3) Kabul	(4) Mazar
Mean				
Alternative-specific constant (alt B)	-0.219*** (0.032)	-0.184*** (0.029)	-0.136*** (0.035)	-0.194*** (0.033)
consXkab	0.060 (0.049)			
consXmaz	0.077* (0.041)			
Transfer: 83% of max (17% less)	-0.640*** (0.036)	-0.211*** (0.051)	-0.834*** (0.075)	-1.001*** (0.066)
Transfer: 67% of max (33% less)	-1.267*** (0.046)	-0.371*** (0.061)	-1.750*** (0.102)	-1.987*** (0.093)
Transfer: 50% of max (50% less)	-1.989*** (0.062)	-0.660*** (0.076)	-2.707*** (0.133)	-3.067*** (0.133)
Frequency: bi-monthly	-0.286*** (0.032)	0.151*** (0.037)	-0.254*** (0.071)	-0.848*** (0.074)
Modality: cash in hand	1.159*** (0.041)	2.244*** (0.091)	0.104* (0.059)	0.869*** (0.067)
Modality: mobile money	-0.641*** (0.038)	-0.410*** (0.060)	-0.031 (0.057)	-1.408*** (0.082)
Recipient: female	0.020 (0.030)	0.635*** (0.046)	-0.327*** (0.070)	-0.430*** (0.051)
Targeting: community-based	-0.923*** (0.035)	-0.929*** (0.052)	-0.395*** (0.055)	-1.405*** (0.082)
SD				
Transfer: 83% of max (17% less)	0.027 (0.071)	0.064 (0.056)	0.089 (0.206)	0.328 (0.350)
Transfer: 67% of max (33% less)	0.209 (0.176)	0.073 (0.092)	0.855*** (0.134)	0.494* (0.269)
Transfer: 50% of max (50% less)	1.687*** (0.076)	1.036*** (0.148)	1.521*** (0.152)	2.278*** (0.157)
Frequency: bi-monthly	1.898*** (0.052)	0.237 (0.190)	2.285*** (0.106)	2.612*** (0.101)
Modality: cash in hand	1.721*** (0.060)	2.430*** (0.107)	0.763*** (0.156)	1.609*** (0.136)
Modality: mobile money	1.829*** (0.061)	2.009*** (0.095)	0.283 (0.379)	2.330*** (0.141)
Recipient: female	1.840*** (0.050)	1.551*** (0.068)	2.254*** (0.118)	1.663*** (0.106)
Targeting: community-based	2.013*** (0.051)	1.767*** (0.076)	1.388*** (0.098)	2.741*** (0.117)
Chi-sq	110.352	41.520	26.222	51.504

Table A6: Test of equal preferences across sites

	χ^2	d.f.	p
Joint tests			
Kabul = DRC	898.6	8	<0.001
Mazar = DRC	830.3	8	<0.001
All sites equal	1219.2	16	<0.001
By attribute $H_0: Kabul = Mazar = 0$			
Transfer: 83% of max	86.4	2	<0.001
Transfer: 67% of max	235.8	2	<0.001
Transfer: 50% of max	295.5	2	<0.001
Bi-monthly delivery	161.4	2	<0.001
Cash in hand	486.4	2	<0.001
Mobile money	147.0	2	<0.001
Female recipient	286.5	2	<0.001
Community targeting	104.5	2	<0.001

Notes: Wald tests on the site $\times$ attribute interactions of the pooled mixed logit (M2), estimated on all 119,352 choice occasions with the DRC as the omitted site. The joint tests ask whether all of a site's interaction terms are jointly zero; the by-attribute tests ask whether a given attribute's two interactions are jointly zero. A rejection indicates that the mean coefficient differs across sites, which is a difference in preferences only if the scale of the error term is common across sites.

Table A7: Willingness to pay by respondent subgroup (1 of 2)

	Food insecure		rCSI high		LCSI high		Head educ. low		Respondent older	
	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Bi-monthly delivery	5.17 (3.88)	17.35*** (4.59)	8.02 (5.12)	15.02*** (3.73)	7.37 (6.10)	13.09*** (3.56)	11.21* (6.07)	13.22*** (3.54)	11.34*** (4.07)	13.57*** (4.62)
Cash in hand	98.31*** (16.97)	204.00*** (31.10)	192.82*** (36.67)	147.32*** (19.74)	165.81*** (39.96)	160.20*** (20.48)	107.63*** (27.35)	182.25*** (22.68)	157.22*** (24.34)	167.96*** (26.81)
Mobile money	-20.83*** (7.07)	-34.66*** (7.30)	-43.35*** (10.40)	-23.93*** (5.86)	-2.49 (9.06)	-36.56*** (6.14)	23.08** (10.05)	-35.99*** (6.31)	-38.25*** (8.10)	-22.50*** (6.75)
Female recipient	23.76*** (5.73)	58.45*** (9.03)	54.21*** (10.69)	38.41*** (5.69)	42.41*** (10.90)	44.93*** (5.96)	37.76*** (10.58)	48.89*** (6.47)	39.47*** (6.70)	49.90*** (8.59)
Community targeting	-27.50*** (6.84)	-88.05*** (13.80)	-80.90*** (16.19)	-60.53*** (8.51)	-73.61*** (18.87)	-62.98*** (8.47)	-52.37*** (13.93)	-73.79*** (9.73)	-63.34*** (10.55)	-70.18*** (11.99)
Choice occasions	6,816	34,704	17,748	23,772	8,136	33,384	3,720	37,740	21,336	20,184

Notes: Willingness to pay in percent of the full transfer forgone, from mixed logits estimated separately within each subgroup arm on the DRC sample, with WTP formed as the ratio of each attribute coefficient to the marginal utility of transfer value. Reference categories are monthly delivery, in-kind food, a male recipient, and data-driven targeting. Each subgroup is a binary split defined in Section 5.4; N is the number of choice occasions in that arm. Standard errors, in parentheses, are from the delta method and clustered at the respondent level; 100 Halton draws. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Willingness to pay by respondent subgroup (2 of 2)

	Wellbeing low		Generalized trust		Received aid		Agency high	
	No	Yes	No	Yes	No	Yes	No	Yes
Bi-monthly delivery	11.53*** (3.28)	19.07* (11.02)	12.31*** (3.54)	7.13 (7.71)	10.23 (8.54)	14.66*** (3.22)	19.91*** (7.03)	9.31*** (3.35)
Cash in hand	149.05*** (17.45)	274.28*** (102.51)	133.52*** (17.30)	296.47*** (85.93)	30.61* (16.73)	163.69*** (18.65)	208.99*** (44.51)	143.75*** (18.82)
Mobile money	-37.31*** (5.78)	6.12 (12.27)	-27.09*** (5.59)	-24.03* (13.37)	-3.13 (12.51)	-27.29*** (5.06)	-74.46*** (16.79)	-8.83* (4.89)
Female recipient	44.58*** (5.48)	56.42** (22.71)	48.51*** (6.57)	39.98*** (12.96)	25.88* (13.42)	43.99*** (5.25)	70.06*** (14.71)	33.78*** (5.19)
Community targeting	-51.69*** (6.54)	-150.88*** (57.32)	-45.42*** (6.53)	-157.47*** (46.02)	-100.73*** (38.48)	-59.67*** (7.42)	-98.69*** (21.19)	-51.97*** (8.00)
Choice occasions	31,200	10,320	24,120	13,608	1,596	37,608	20,988	20,532

Notes: Willingness to pay in percent of the full transfer forgone, from mixed logits estimated separately within each subgroup arm on the DRC sample, with WTP formed as the ratio of each attribute coefficient to the marginal utility of transfer value. Reference categories are monthly delivery, in-kind food, a male recipient, and data-driven targeting. Each subgroup is a binary split defined in Section 5.4; N is the number of choice occasions in that arm. Standard errors, in parentheses, are from the delta method and clustered at the respondent level; 100 Halton draws. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Willingness to pay by subgroup and site

	DRC		Kabul		Mazar-i-Sharif	
	No	Yes	No	Yes	No	Yes
Food insecure						
Bi-monthly delivery	5.17 (3.88)	17.35*** (4.59)	-5.27*** (1.54)	-3.17 (2.14)	-18.30*** (1.75)	-7.04*** (1.49)
Cash in hand	98.31*** (16.97)	204.00*** (31.10)	2.38* (1.27)	2.18 (1.91)	17.39*** (1.46)	10.47*** (1.59)
Mobile money	-20.83*** (7.07)	-34.66*** (7.30)	-0.62 (1.16)	-1.96 (2.14)	-27.78*** (1.83)	-13.93*** (1.72)
Female recipient	23.76*** (5.73)	58.45*** (9.03)	-9.80*** (1.50)	2.29 (2.24)	-5.97*** (1.06)	-10.87*** (1.52)
Community targeting	-27.50*** (6.84)	-88.05*** (13.80)	-6.94*** (1.08)	-9.81*** (2.27)	-23.83*** (1.66)	-19.23*** (1.85)
Choice occasions	6,816	34,704	22,008	4,320	34,992	16,512
High coping index (rCSI)						
Bi-monthly delivery	8.02 (5.12)	15.02*** (3.73)	-0.80 (1.53)	-7.83*** (2.05)	-18.80*** (1.51)	-0.80 (1.27)
Cash in hand	192.82*** (36.67)	147.32*** (19.74)	2.62* (1.46)	1.96 (1.53)	16.32*** (1.31)	11.50*** (1.79)
Mobile money	-43.35*** (10.40)	-23.93*** (5.86)	-0.48 (1.42)	-0.78 (1.44)	-25.38*** (1.63)	-16.21*** (2.04)
Female recipient	54.21*** (10.69)	38.41*** (5.69)	-9.93*** (1.84)	-2.60 (1.63)	-8.61*** (1.04)	-4.60*** (1.31)
Community targeting	-80.90*** (16.19)	-60.53*** (8.51)	-7.15*** (1.34)	-8.06*** (1.45)	-17.54*** (1.31)	-31.40*** (2.81)
Choice occasions	17,748	23,772	13,752	12,576	39,276	12,228
Female respondent						
Bi-monthly delivery			-4.93** (1.97)	-4.76*** (1.64)	-10.80*** (1.20)	-24.26*** (3.08)
Cash in hand			2.43* (1.45)	2.21 (1.62)	15.38*** (1.18)	9.75*** (2.44)
Mobile money			-0.87 (1.39)	-1.35 (1.52)	-20.79*** (1.39)	-26.90*** (3.33)
Female recipient			-16.16*** (1.91)	5.75*** (1.67)	-9.80*** (0.91)	2.37 (1.85)
Community targeting			-7.38***	-8.67***	-17.27***	-37.37***

Table A10: Willingness to pay translated into local currency, by site

	DRC (USD)	Kabul (AFN)	Mazar-i-Sharif (AFN)
Frequency <i>Ref.: monthly</i>			
Bi-monthly	1.78*** (0.41)	-136.93*** (37.89)	-424.77*** (36.10)
Modality <i>Ref.: site-specific third option</i>			
Cash in hand	21.68*** (2.45)	74.18** (32.02)	440.84*** (32.51)
Mobile money	-3.91*** (0.68)	-28.81 (30.84)	-690.38*** (39.46)
Recipient <i>Ref.: male</i>			
Female	6.13*** (0.73)	-203.53*** (37.14)	-234.15*** (26.02)
Targeting <i>Ref.: data-driven</i>			
Community-based	-8.82*** (1.07)	-222.68*** (29.56)	-666.43*** (35.68)
Full transfer	12.82 USD	3,000 AFN	3,000 AFN
Observations	41,520	26,328	51,504
Respondents	3,460	2,194	4,292

Each cell translates the percentage WTP estimate of Table 5 into money by multiplying it by the site's full transfer. Point estimates are followed by SEs in parentheses.

The DRC column is in nominal USD, converted from 30,000 CDF at the Sept–Oct 2023 exchange rate of 2,340 CDF/USD. The Afghan columns are reported in AFN rather than USD: the World Bank publishes no Afghanistan exchange rate after 2020, so there is no independent official rate for the Jan–Feb 2025 survey period. At the operational rate implied by the instrument (71.43 AFN/USD, from presenting 3,000 AFN as \$42), 3,000 AFN is about \$42.00.

The DRC transfer is denominated per household member and the Afghan transfer per household, so money amounts are not comparable across sites; Table 5 is the comparable metric.

SEs (delta method) clustered at the respondent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Preferences for targeting-committee design

	Coef. (1)	SD (2)
Composition <i>Ref.: anyone can join</i>		
Elected	0.25*** (0.04)	1.12*** (0.06)
Community leaders	-0.13*** (0.04)	0.80*** (0.06)
Gender <i>Ref.: mixed</i>		
All-female	-0.27*** (0.03)	0.64*** (0.08)
All-male	-0.76*** (0.04)	0.92*** (0.06)
Process <i>Ref.: public</i>		
Private	-0.96*** (0.05)	1.97*** (0.06)
Observations	41,520	
No. respondents	3,460	
LR chi2	773.4	
Log-likelihood	-12031	
Halton draws	100	

Mixed-logit estimates. Coefficients are interpreted relative to the indicated reference category. SD columns capture preference heterogeneity. SEs clustered at the respondent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Committee preferences by respondent subgroup (1 of 2)

	Food insecure		rCSI high		LCSI high		Head educ. low		Respondent older	
	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Composition: elected	0.34*** (0.11)	0.24*** (0.04)	0.10 (0.06)	0.37*** (0.05)	0.67*** (0.10)	0.16*** (0.04)	0.49*** (0.14)	0.23*** (0.04)	0.20*** (0.05)	0.30*** (0.06)
Composition: leaders	-0.12 (0.11)	-0.14*** (0.04)	-0.12** (0.06)	-0.14*** (0.05)	-0.23*** (0.09)	-0.11*** (0.04)	-0.12 (0.13)	-0.13*** (0.04)	-0.15*** (0.05)	-0.12** (0.05)
Gender: all female	-0.33*** (0.09)	-0.25*** (0.04)	-0.32*** (0.05)	-0.21*** (0.05)	-0.45*** (0.08)	-0.23*** (0.04)	-0.45*** (0.12)	-0.25*** (0.04)	-0.28*** (0.05)	-0.26*** (0.05)
Gender: all male	-0.77*** (0.12)	-0.75*** (0.04)	-0.85*** (0.07)	-0.67*** (0.05)	-1.03*** (0.11)	-0.69*** (0.04)	-0.88*** (0.16)	-0.73*** (0.04)	-0.72*** (0.06)	-0.80*** (0.06)
Process: private	-1.38*** (0.14)	-0.88*** (0.05)	-0.72*** (0.07)	-1.15*** (0.07)	-0.93*** (0.10)	-0.96*** (0.05)	-1.20*** (0.18)	-0.92*** (0.05)	-0.85*** (0.06)	-1.07*** (0.07)
Choice occasions	6,816	34,704	17,748	23,772	8,136	33,384	3,720	37,740	21,336	20,184

Notes: Mean mixed-logit coefficients from the targeting-committee experiment, estimated separately within each subgroup arm on the DRC sample. Reference categories are a committee open to anyone, mixed gender, and public deliberation. The committee design contains no monetary attribute, so these are utility coefficients rather than willingness to pay: they are identified only up to each arm's own error scale, and should be compared within a column rather than across columns. Each subgroup is a binary split defined in Section 5.4; N is the number of choice occasions in that arm. Standard errors in parentheses, clustered at the respondent level; 100 Halton draws. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Committee preferences by respondent subgroup (2 of 2)

	Wellbeing low		Generalized trust		Received aid		Agency high		Randomized to CBT	
	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Composition: elected	0.34*** (0.05)	-0.10 (0.08)	0.26*** (0.05)	0.37*** (0.08)	0.26 (0.21)	0.25*** (0.04)	0.10* (0.05)	0.42*** (0.06)	0.19*** (0.05)	0.31*** (0.06)
Composition: leaders	-0.08** (0.04)	-0.29*** (0.06)	-0.08* (0.04)	-0.19*** (0.07)	0.15 (0.16)	-0.16*** (0.04)	-0.17*** (0.05)	-0.10* (0.06)	-0.15*** (0.05)	-0.12** (0.05)
Gender: all female	-0.28*** (0.04)	-0.23*** (0.07)	-0.20*** (0.05)	-0.40*** (0.07)	-0.56** (0.22)	-0.28*** (0.04)	-0.23*** (0.05)	-0.31*** (0.05)	-0.25*** (0.05)	-0.29*** (0.05)
Gender: all male	-0.80*** (0.05)	-0.59*** (0.08)	-0.74*** (0.05)	-0.75*** (0.08)	-0.33 (0.21)	-0.74*** (0.04)	-0.72*** (0.06)	-0.77*** (0.06)	-0.70*** (0.05)	-0.81*** (0.06)
Process: private	-0.92*** (0.05)	-1.15*** (0.12)	-0.81*** (0.06)	-1.37*** (0.10)	-1.54*** (0.40)	-0.93*** (0.05)	-0.76*** (0.06)	-1.16*** (0.07)	-0.93*** (0.06)	-0.99*** (0.07)
Choice occasions	31,200	10,320	24,120	13,608	1,596	37,608	20,988	20,532	21,036	20,484

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Notes: Mean mixed-logit coefficients from the targeting-committee experiment, estimated separately within each subgroup arm on the DRC sample. Reference categories are a committee open to anyone, mixed gender, and public deliberation. The committee design contains no monetary attribute, so these are utility coefficients rather than willingness to pay: they are identified only up to each arm's own error scale, and should be compared within a column rather than across columns. Each subgroup is a binary split defined in Section 5.4; N is the number of choice occasions in that arm. Standard errors in parentheses, clustered at the respondent level; 100 Halton draws. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Welfare-equivalent transfer change under aid-cut and redesign scenarios

Scenario	Welfare-equiv. change (% of full transfer)
S0: Baseline (full, monthly, in-kind, male, data)	0.0
S1: Cut 17% only	-17.2
S2: Cut 33% only	-30.2
S3: Cut 50% only	-53.7
S4: Cut 33% + cash in hand	152.5
S5: Cut 33% + mobile money	-63.6
S6: Cut 33% + bi-monthly	-17.9
S7: Cut 33% + female recipient	21.5
S8: Cut 33% + community targeting	-105.9
S9: Cut 33% + cash + female + bi-monthly	216.5
S10: Cut 50% + cash + female + bi-monthly	193.0

Welfare-equivalent change from the baseline package (full transfer, monthly delivery, in-kind modality, male recipient, data-driven targeting), in percent of the full transfer. Each value translates the indirect-utility difference between the scenario and the baseline into transfer-value units using the average marginal utility of cash; positive values mean the scenario is preferred to the baseline. Estimates use the mixed-logit mean coefficients.

Cross-language replication. As an independent check on the estimation, the pooled and by-site preference models were re-estimated in R with the `apollo` package—a different mixed-logit implementation, optimiser, and simulation-draw generator. The two engines reproduce the Stata estimates: across all 70 parameters the largest standardised difference is 0.83 standard errors, and the pooled mean coefficients agree to within 3.9% (Table A19). The residual gaps are the expected Monte-Carlo noise from the two draw generators (both use 100 draws: Halton in Stata, MLHS in R), not substantive disagreement.

Table A19: Cross-language check of the pooled preference model: Stata `mixlogit` vs. R `apollo`

Parameter	Stata	Apollo	% Δ	z
<i>Pooled (site ASCs)</i>				
ASC (alt B)	-0.219	-0.224	2.5	0.12

continued on next page

Table A19 (continued)

Parameter	Stata	Apollo	% Δ	z
$\times$ Kabul	0.060	0.045	24.3	0.21
$\times$ Mazar	0.077	0.078	-1.0	-0.01
Transfer 83% of max (mean)	-0.640	-0.655	2.4	0.30
Transfer 67% of max (mean)	-1.267	-1.316	3.9	0.74
Transfer 50% of max (mean)	-1.989	-2.041	2.6	0.58
Bi-monthly delivery (mean)	-0.286	-0.325	13.5	0.83
Cash in hand (mean)	1.159	1.201	-3.6	-0.70
Mobile money (mean)	-0.641	-0.660	2.9	0.35
Female recipient (mean)	0.020	0.024	—	-0.09
Community targeting (mean)	-0.923	-0.937	1.5	0.28
Transfer 83% of max (SD)	0.027	0.059	—	-0.36
Transfer 67% of max (SD)	0.209	0.099	52.4	0.54
Transfer 50% of max (SD)	1.687	1.624	3.8	0.53
Bi-monthly delivery (SD)	1.898	1.958	-3.2	-0.78
Cash in hand (SD)	1.721	1.854	-7.8	-1.51
Mobile money (SD)	1.829	1.863	-1.9	-0.39
Female recipient (SD)	1.840	1.931	-5.0	-1.25
Community targeting (SD)	2.013	2.124	-5.6	-1.49
<i>DRC</i>				
ASC (alt B)	-0.184	-0.193	4.8	0.21
Transfer 83% of max (mean)	-0.211	-0.223	5.8	0.16
Transfer 67% of max (mean)	-0.371	-0.424	14.3	0.60
Transfer 50% of max (mean)	-0.660	-0.742	12.4	0.75
Bi-monthly delivery (mean)	0.151	0.161	-6.3	-0.18
Cash in hand (mean)	2.244	2.303	-2.6	-0.45
Mobile money (mean)	-0.410	-0.446	8.7	0.41
Female recipient (mean)	0.635	0.645	-1.5	-0.15
Community targeting (mean)	-0.929	-0.946	1.9	0.23
Transfer 83% of max (SD)	0.064	0.000	99.4	0.79

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Table A19 (continued)

Parameter	Stata	Apollo	% Δ	z
Transfer 67% of max (SD)	0.073	0.142	-94.7	-0.53
Transfer 50% of max (SD)	1.036	0.852	17.7	0.71
Bi-monthly delivery (SD)	0.237	0.411	-73.9	-0.78
Cash in hand (SD)	2.430	2.516	-3.5	-0.56
Mobile money (SD)	2.009	2.101	-4.6	-0.66
Female recipient (SD)	1.551	1.708	-10.2	-1.53
Community targeting (SD)	1.767	1.864	-5.4	-0.85
<i>Kabul</i>				
ASC (alt B)	-0.136	-0.140	3.0	0.08
Transfer 83% of max (mean)	-0.834	-0.876	5.1	0.40
Transfer 67% of max (mean)	-1.750	-1.753	0.2	0.02
Transfer 50% of max (mean)	-2.707	-2.765	2.1	0.30
Bi-monthly delivery (mean)	-0.254	-0.205	-19.3	-0.49
Cash in hand (mean)	0.104	0.137	-32.6	-0.40
Mobile money (mean)	-0.031	-0.033	—	0.03
Female recipient (mean)	-0.327	-0.323	-1.3	-0.04
Community targeting (mean)	-0.395	-0.384	-2.7	-0.14
Transfer 83% of max (SD)	0.089	0.149	-68.1	-0.21
Transfer 67% of max (SD)	0.855	0.773	9.5	0.31
Transfer 50% of max (SD)	1.521	1.690	-11.2	-0.67
Bi-monthly delivery (SD)	2.285	2.415	-5.7	-0.82
Cash in hand (SD)	0.763	0.746	2.3	0.08
Mobile money (SD)	0.283	0.185	34.6	0.23
Female recipient (SD)	2.254	2.294	-1.8	-0.24
Community targeting (SD)	1.388	1.475	-6.3	-0.60
<i>Mazar</i>				
ASC (alt B)	-0.194	-0.195	0.3	0.01
Transfer 83% of max (mean)	-1.001	-1.058	5.7	0.61

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Table A19 (continued)

Parameter	Stata	Apollo	% Δ	z
Transfer 67% of max (mean)	-1.987	-2.070	4.2	0.63
Transfer 50% of max (mean)	-3.067	-3.149	2.7	0.44
Bi-monthly delivery (mean)	-0.848	-0.881	4.0	0.33
Cash in hand (mean)	0.869	0.849	2.3	0.21
Mobile money (mean)	-1.408	-1.386	-1.6	-0.19
Female recipient (mean)	-0.430	-0.423	-1.5	-0.09
Community targeting (mean)	-1.405	-1.411	0.4	0.06
Transfer 83% of max (SD)	0.328	0.031	90.7	0.75
Transfer 67% of max (SD)	0.494	0.141	71.4	0.82
Transfer 50% of max (SD)	2.278	2.131	6.4	0.65
Bi-monthly delivery (SD)	2.612	2.657	-1.7	-0.30
Cash in hand (SD)	1.609	1.652	-2.7	-0.24
Mobile money (SD)	2.330	2.404	-3.2	-0.37
Female recipient (SD)	1.663	1.793	-7.8	-0.91
Community targeting (SD)	2.741	2.847	-3.9	-0.64

Same preference-space mixed logit fitted by two engines on the identical pooled choice data. Stata uses 100 Halton draws; Apollo uses 100 MLHS draws, so agreement is expected within a fraction of a Monte-Carlo standard error. SD is reported as a magnitude (its sign is not identified). % Δ omitted where the coefficient is within 0.05 of zero. $z = (b_{\text{Stata}} - b_{\text{Apollo}}) / \sqrt{se_{\text{Stata}}^2 + se_{\text{Apollo}}^2}$.

Table A15: Welfare recovery from compensating design switches under a 33% transfer cut

Compensating switch	DRC	
	ΔV	% rec.
Cash in hand	2.244	+604.4%
Mobile money	-0.410	-110.5%
Bi-monthly delivery	0.151	+40.8%
Female recipient	0.635	+171.0%
Community targeting	-0.929	-250.3%
<i>Reference: 33% cut</i>	-0.371	100%

Notes: ΔV is the change in indirect utility from a one-step change in the indicated attribute relative to its reference category. Recovery is computed as $\Delta V/|V_{cut33}| \times 100$, where V_{cut33} is the utility loss from a 33% transfer cut. Positive recovery means the switch offsets part of the welfare loss; values above 100% imply full compensation. Estimates use the DRC mixed logit.

Table A16: Household characteristics and the choice of a more liquid package

	(1) DRC + Mazar	(2) Kabul
chose_liquid		
Food insecure	-0.025 (0.026)	0.032 (0.048)
High coping (rCSI)	0.023 (0.026)	-0.041 (0.036)
Female respondent	-0.215*** (0.032)	-0.014 (0.038)
Older respondent	0.006 (0.023)	0.003 (0.036)
Low education	0.046 (0.041)	0.044 (0.041)
Free mobility	-0.020 (0.029)	0.040 (0.048)
High agency	0.115*** (0.025)	-0.132*** (0.040)
Choice occasions	43,020	13,110
Respondents	7,170	2,185
Pseudo- R^2	0.073	0.086

Notes: Logit estimates. The unit is the choice occasion, reduced to a single contrast: the dependent variable is one if the respondent selected the more liquid of the two packages offered. Modalities are ranked cash in hand > mobile money > the site-specific third option. That ranking is straightforward in the DRC, where the third option is in-kind food, and in Mazar, where it is a voucher; it is *not* straightforward in Kabul, where the third option is a bank deposit that may be more liquid than a mobile balance, so Kabul is reported separately in column (2) rather than pooled into column (1). Occasions on which both packages carried the same modality offer no liquidity contrast and are dropped. All models control for the differences between the two packages in transfer value, delivery frequency, recipient, and targeting method, so the coefficients describe who trades those features away for liquidity; column (1) also includes a site fixed effect. Because every DRC respondent is female, the respondent-sex coefficient in column (1) is identified within Mazar alone and should not be read as a DRC result. Standard errors, in parentheses, are clustered at the respondent level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Transfer-size disutility — dummy, linear, and quadratic mixed logits

	Dummy (1)	Linear (2)	Quadratic (3)
Transfer: 17% less	-0.213***		
Transfer: 33% less	-0.399***		
Transfer: 50% less	-0.703***		
Transfer cut (% foregone)		-0.014***	-0.010***
Transfer cut squared			-0.000
Log-likelihood	-10856.5	-10857.3	-10856.8
AIC	21741.0	21738.7	21739.5
BIC	21861.9	21842.3	21851.7

Mixed-logit estimates (DRC sample) differing only in how transfer size enters; all share the same random-coefficient structure on the non-price attributes (frequency, modality, recipient, targeting). Transfer terms enter as fixed coefficients, so the linear specification is nested in the dummy specification. Likelihood-ratio test of linear vs. dummy: $\chi^2(2) = 1.67$, $p = 0.433$. SEs clustered at the respondent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A18: Experimental design checks

	Share of profiles	Max. $ \rho $ with another attribute	Overlap rate
Panel A. Transfer-attribute experiment			
Transfer: 83% of max	0.250	0.066	0.500
Transfer: 67% of max	0.249	0.069	0.502
Transfer: 50% of max	0.251	0.071	0.499
Bi-monthly delivery	0.500	0.003	0.000
Cash in hand	0.333	0.069	0.335
Mobile money	0.334	0.071	0.332
Female recipient	0.500	0.005	0.000
Community-based targeting	0.500	0.004	0.000
Panel B. Targeting-committee experiment			
Composition: elected	0.334	0.030	0.332
Composition: leaders	0.334	0.060	0.332
Gender: all female	0.333	0.060	0.335
Gender: all male	0.334	0.032	0.332
Process: private	0.500	0.002	0.000

Share of profiles is the fraction of alternatives carrying the level, so a balanced design puts it at 1/2 for a two-level attribute and 1/3 for a three-level one.

Max. $|\rho|$ with another attribute is the largest absolute pairwise correlation between the level and any level of a *different* attribute. Correlations within an attribute are omitted because they are mechanical: exactly one level of each attribute appears on every profile, which forces its dummies to be negatively correlated (about -0.33 among the three transfer levels, -0.50 between the two modality levels). Independence across attributes is what identification requires, and is what this column reports.

Overlap rate is the share of choice sets in which both alternatives carry the level. A rate of zero means the attribute always differs between the two profiles, so every task is informative about it; higher rates trade statistical efficiency for cognitive realism.

Both designs are D-optimal; efficiency metrics are in Table A2.